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Analytical Tomographic 3D Reconstruction in Medical Imaging

Implementation of the Katsevich Inversion Formula in an Open-Source Python Framework

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Abstract

Computed tomography (CT) is a key imaging technique in medicine and industry. In 3D cone-beam CT, however, the standard filtered backprojection (FBP) algorithm is only approximate and produces artifacts, especially in helical high-pitch scenarios. In 2002, Katsevich derived the first exact inversion formula for helical cone-beam CT. Despite this breakthrough, existing implementations are incomplete, proprietary, or limited to simplified setups, which has limited its adoption.

This thesis presents an open-source implementation of the Katsevich algorithm within the Operator Discretization Library (ODL). The focus is on supporting realistic geometries, including flat and curved detectors, variable pitch, and flying focal spot (FFS). The algorithm was structured into modular steps: weighting, rebinning, filtering, and backprojection. Existing ODL and ASTRA operators were reused where possible, while new components were developed when necessary, such as GPU-accelerated voxel-dependent weighting and flexible computation of Tam–Danielsson (TD) windows.

The method was validated on synthetic phantoms and clinical datasets. Katsevich reconstructions show improved structural preservation and fewer helical artifacts along the rotation axis compared to FBP. Remaining limitations include imperfect curved-detector modeling, partial handling of FFS, and high computational cost in high-resolution variable-pitch scans.

The main contribution is to make the Katsevich algorithm openly available in ODL, providing a foundation for reproducibility and for further research.

Keywords

Katsevich Algorithm, Exact Analytical Reconstruction, Helical Cone-Beam CT, 3D CT Reconstruction, Filtered Backprojection (FBP), Weighted Backprojection, Variable Pitch CT, Flying Focal Spot (FFS), Implementation of Katsevich Reconstruction, Tam-Danielsson Window, Curved Detector Geometry, Clinical Data

Sammanfattning

Datortomografi (CT) är en viktig bildteknik inom både medicin och industri. I tredimensionell kon-beam CT används ofta filtrerad bakprojektion (FBP), som endast är approximativ och kan ge upphov till artefakter, särskilt i spiralformade högpitchenscenarier. År 2002 presenterade Katsevich den första exakta inversionsformeln för spiral-CT. Trots detta teoretiska genombrott har befintliga implementationer varit begränsade, proprietära eller anpassade till förenklade scenarier, vilket har bromsat metodens användning.

I detta examensarbete utvecklas en öppen implementation av Katsevich-algoritmen inom biblioteket Operator Discretization Library (ODL). Fokus har legat på att stödja realistiska geometrier, inklusive plana och böjda detektorer, variabel pitch och rörligt fokus. Algoritmen har delats upp i modulära steg (viktning, rebinning, filtrering och bakprojektion) där befintliga komponenter återanvänds när det är möjligt. Nya delar utvecklas vid behov, exempelvis GPU-accelererad voxelberoende viktning och flexibel beräkning av Tam–Danielsson-fönster.

Metoden har validerats på syntetiska fantom och kliniska dataset. Resultaten visar att Katsevich ger bättre bevarande av strukturer och färre artefakter längs rotationsaxeln jämfört med filtrerad bakprojektion. Begränsningar kvarstår, såsom ofullständig modellering av böjda detektorer, endast delvis hantering av rörligt fokus och hög beräkningskostnad vid hög upplösning och variabel pitch.

Huvudbidraget är att göra Katsevich-algoritmen öppet tillgänglig, vilket möjliggör reproducerbarhet och lägger grunden för framtida forskning och förbättringar.

Nyckelord

Katsevich-algoritmen, Exakt analytisk rekonstruktion, Spiralformad kon-beam CT, 3D CT-rekonstruktion, Filtrerad bakprojektion (FBP), Viktad bakprojektion, CT med variabel pitch, Rörligt fokus (FFS), Implementation av Katsevich-rekonstruktion, Tam–Danielsson-fönster, Böjd detektorgeometri, Kliniska data

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I am definitely proud of myself for completing this project, not just for the academic achievement it represents, but for the perseverance it required. There were times when self-doubt and impostor syndrome made me question whether I truly belonged in this master's program. Now, standing on the other side, I know that I do, and I know that this version of myself could not have come into being without the people that shaped this journey.

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This thesis reflects a lot of work in studying applied mathematics, but it also reflects growth, community, and perseverance. And for all of that, I'm thankful.

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Chapter 1

Introduction

Computed tomography (CT) is a fundamental imaging technique used to reconstruct 2D or 3D representations of internal structures from a series of x-ray projections. In 2D, reconstruction is relatively straightforward: the Fourier Slice Theorem provides a clear relationship between projection data and the image's frequency content. This allows to build efficient analytical algorithms such as filtered backprojection (FBP):

$$f = \mathcal{R}^*(v * g),$$

where g is the projected measured data, f is the image to be reconstructed, $\mathcal{R}^*$ is the adjoint to the Radon Transform and v is a convolution kernel. Due to its speed and accuracy, algorithms of the FBP-type became the standard approach.

However, when extending CT to 3D, the problem becomes significantly more complex. In particular, for cone-beam CT, the standard geometry used in modern scanners, there is no relation between projection data and planar slices of the object's Fourier transform. As a result, the Fourier Slice Theorem is not directly applicable to cone-beam CT, and the projection data cannot be mapped straightforwardly to the object's Fourier space.

Before the development of analytically exact 3D algorithms, most reconstruction approaches fell into two broad categories: approximate analytical methods and iterative methods. Among the approximate methods, the most widely used is the Feldkamp–Davis–Kress (FDK) algorithm [1], an extension of FBP to circular cone-beam geometry. Although FDK is computationally efficient and remains widely used in practice, it is not analytically exact and it introduces significant cone-beam artifacts, particularly for geometries using a large angle in the X-rays cone beam [2].

On the other hand iterative reconstruction methods, such as the Algebraic Reconstruction Technique (ART) [3] and the Statistical Iterative Reconstruction (SIR), treat the reconstruction as a linear inverse problem. That is, the image is reconstructed by solving a system of linear equations:

$$g = Af + e,$$

where f is the image to be reconstructed, g is the measured projection data, A the system matrix determined by the CT system geometry, and e the error vector. The error e is minimized in an optimization process, where known information about the image f, often available, can be incorporated as a penalty term. These methods have been shown to perform better than standard FBP-type algorithms when working with sparse and noisy data, and are able to deliver an equivalent image quality using a low-dose of x-rays [4, 5]. However, they are computationally expensive and often requires careful tuning of regularization parameters. Moreover, the solution may depend heavily on initialization, convergence criteria, and discretization choices, making the results harder to interpret and replicate compared to analytical methods. [2, 6]

Thus, despite continuous improvements in solving reconstruction problems, a major limitation persisted: there was no known reconstruction algorithm that could be theoretically exact for 3D cone-beam CT.

This changed with the work of Katsevich, who proposed in 2002 the first theoretically exact inversion formula for helical cone-beam CT [7]. His algorithm was a major breakthrough because it provided a filtered backprojection-type method that was analytically exact in the continuous setting for helical trajectories. This result opened the door to new developments in 3D reconstruction . Following the original work, Katsevich and others extended the approach to a wider class of geometries, including nonstandard helical trajectories, variable pitch, and more general source curves. [8, 9]

However, practical use of the Katsevich algorithm remains limited. CT data are always sampled discretely, and any real implementation must deal with numerical challenges and errors given by interpolation, noise, and discretization. Furthermore, while the mathematical structure of the algorithm is known, up to our knowledge no implementation is publicly available. Some exist only in proprietary industrial software; others are described in academic papers without accessible or runnable code.

This has created quite a fragmented research landscape. Many studies on Katsevich-type algorithms focus on simplified acquisition setups (*e.g.*, flat detector and with constant pitch), and implementations are not available. [10–13]. As a result, it is difficult to reproduce results, compare methods across variants, and in general explore the full potential of the algorithm in realistic CT scenarios.

The aim of this thesis is therefore:

1. To implement a flexible and correct version of the Katsevich reconstruction algorithm that supports realistic acquisition geometries, including:
   - both flat and curved detectors,
   - constant and variable pitch trajectories,
   - flying focal spot (FFS).
2. To integrate this implementation into the open-source Python package ODL, used for prototyping and solving inverse problems, especially in tomography. This will allow other researchers to easily test, reuse, and build on the implementation.

The implementation developed in this thesis is based on a detailed study of Katsevich's original work, as well as on some follow-up papers proposing generalizations of the acquisition geometry [14], and modular formulations of the algorithm [15, 16]. These include strategies for decomposing the mathematical formulation of the reconstruction into subsequent steps of weighting, rebinning, filtering and backprojecting the projection data. Particular care was taken to design the code in a modular way, allowing to have large flexibility in the acquisition geometry and to reuse existing ODL components, when possible. In addition, small portions of code (related to GPU-accelerated voxel-wise weighting) were adapted from an external source [17].

Initial validation was carried out using synthetic sinogram data obtained using a constant pitch. The implementation was then extended to more realistic scenarios, including curved detectors and data acquisition using a variable pitch and flying focal spot, and compared to ODL's standard FBP operator outputs across different acquisition scenarios.

Note that this work focuses primarily on the correctness of the reconstruction algorithm. Issues of memory optimization and runtime performance were not prioritized. Furthermore, the reconstruction assumes that all the rays in the cone-beam have equal intensity. In practice, some systems apply a correction factor based on the distance from the detector center, but this was not included in our model.

To address the objectives outlined above, this thesis is organized into six chapters. The theoretical foundations of CT image reconstruction are laid out in Chapter 2, which begins with classical approaches such as filtered backprojection and moves to the Katsevich inversion formula. This chapter also introduces the mathematical framework used throughout the work, details the modeling of both flat and curved detector geometries, and presents the ODL and ASTRA software libraries used as basis for the implementation.

Building on this foundation, Chapter 3 presents the overall reconstruction strategy, walking through the derivation of the Katsevich algorithm for both detector types. The chapter also introduces the practical implementation within the ODL framework, summarizing the main design choices and operator structure, while deferring the in-depth code analysis to Appendix D.

The focus then shifts to the experimental framework. In Chapter 4, we specify the quantitative metrics used for performance evaluation and describe the datasets used in the study, including both simulated phantoms and real CT measurements.

Results are presented in Chapter 5, where the Katsevich algorithm is tested on both synthetic phantoms and CT datasets. These experiments assess the quality of the reconstruction, using comparisons either against the ground truth phantom (when available) or standard FBP results.

In Chapter 6, we take a critical look at the current limitations of the implementation, identifying opportunities for software improvements.

Finally, Chapter 7 closes the thesis with a summary of the main contributions and suggests possible directions for future work.

Chapter 2

Background

2.1 Computed Tomography

Computed Tomography (CT) is a non-invasive imaging technique that aims to reconstruct the internal structure of an object from a set of X-ray projections acquired from different directions. It relies on the physical principle that X-rays are attenuated as they pass through matter, and this attenuation is related to the internal composition of the object.

When an X-ray beam travels through a homogeneous material, its intensity decreases exponentially according to the Beer–Lambert law [18]:

$$I = I_0 e^{-\mu \Delta x},$$

where I_0 is the initial intensity, μ is the linear attenuation coefficient of the material, and Δx is the distance traveled by the ray. For an inhomogeneous object where $\mu = \mu(x)$ varies spatially, the measured intensity along a ray path l is given by:

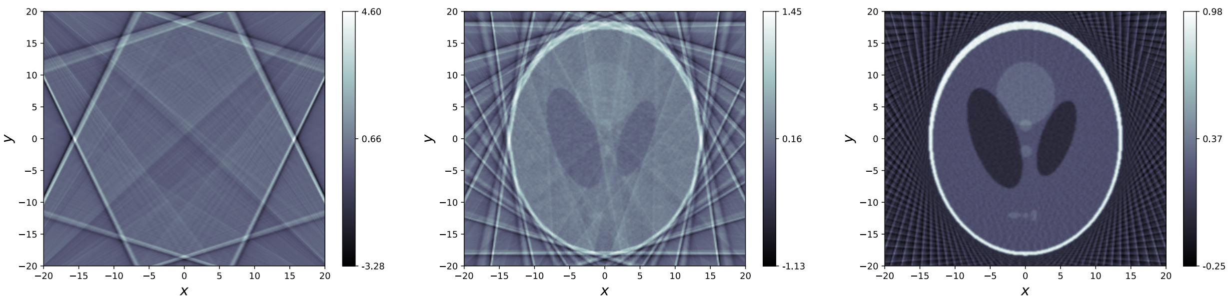
$$I(l) = I_0 \exp\left(-\int_l \mu(x)\, dx\right),$$

which yields the linear relation between the data and the unknown:

$$-\ln\left(\frac{I(l)}{I_0}\right) = \int_l \mu(x)\, dx =: \mathcal{R}\mu(l).$$

This integral is known as the Radon Transform or forward projection, and CT reconstruction corresponds to inverting this transform to recover μ from the set of known integrals $\mathcal{R}\mu(l)$.

Note that acquiring projection data from a single viewpoint leads inevitably to information of the interior being lost. Instead, obtaining projections from various angles allows to capture more information. This concept was formalized in 1917 by Radon [19] who proved that the attenuation of an object $\mu : \mathbb{R}^2 \to \mathbb{R}$ can be recovered exactly from an infinite set of projections. In practice, only a finite number of line integrals can be measured. Clearly, more acquisition angles lead to a better reconstruction.

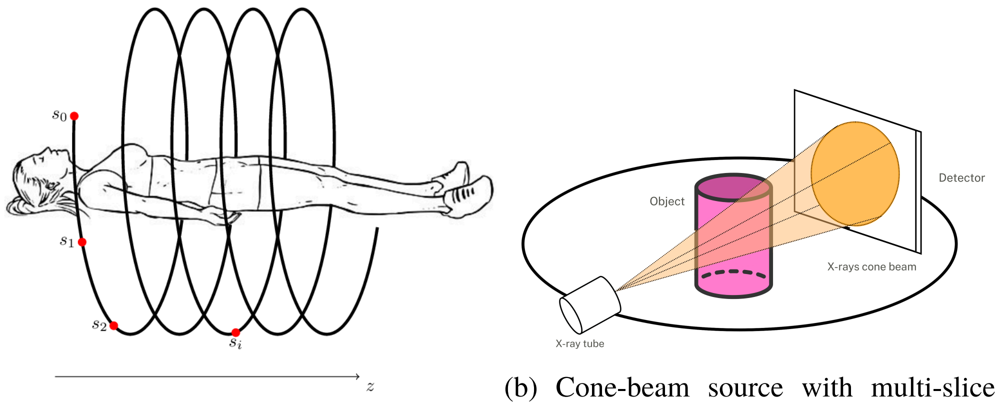


(a) 4 views (b) 20 views (c) 100 views

Figure 2.1: 2D reconstruction of the Shepp-Logan phantom using different amount of acquisition angles. Insufficient views cause distortions in the reconstruction.

It wasn't until 1971 that the first commercial scanner was built, by Godfrey Hounsfield in the EMI Central Research, earning him and and Allan Cormack the Nobel Prize. Their early scanner used a parallel beam geometry, collecting one ray at a time with a linear motion and rotating the system to change the viewpoint of the acquisition. Because this approach was extremely slow, the second generation of CT scanners introduced a linear array of detectors and a source emitting diverging beams (fan-beam or cone-beam geometry), allowing multiple rays to be acquired at once. With this modifications, the scan time got drastically reduced. The third generation changed the acquisition geometry by introducing a curved detector array. In this case the x-rays source and the detector rotate continuously around the object, they do not translate anymore. This improved image resolution and halved the acquisition time. Most modern systems are based on this third-generation geometry. State-of-the-art CT scanners use multi-slice, helical acquisition and flying focal spot (FFS) techniques to increase sampling density and reduce radiation dose, see Figure 2.2.

The mathematical formulation of CT reconstruction is clearest in the 2D parallel-beam case. In this setting, the Radon transform of a function $\mu(x, y)$,



(a) Helical trajectory.

(b) Cone-beam source with multi-slice detector.

Figure 2.2: State of the art techniques in CT scanners.

which represents the internal structure of the object, is defined as the collection of its line integrals. Each projection is taken at an angle θ, and records how much the X-rays are attenuated along lines perpendicular to the direction $a(\theta) = (\cos\theta, \sin\theta)$. The measured projection data is:

$$g(\theta, s) = \int_{\mathbb{R}^2} \mu(x, y)\delta(s - x\cos\theta - y\sin\theta)\,dx\,dy, \tag{2.1}$$

where s is the (signed) distance of the line from the origin. A central theoretical result in this context is the Fourier Slice Theorem. It states that if we take the 1D Fourier transform of $g(\theta, s)$ with respect to s, the result is equal to the 2D Fourier transform of μ evaluated along the line in frequency space passing through the origin in the direction θ:

$$\hat{g}(\theta, \omega) = \hat{\mu}(\omega\cos\theta, \omega\sin\theta). \tag{2.2}$$

In other words, each projection provides a radial slice of the object's 2D Fourier transform. This observation leads to a natural reconstruction strategy: by acquiring projection data at many different angles θ, one obtains values of μ along many radial lines. If these lines densely fill the frequency domain, the collected data samples can be interpolated from polar to Cartesian coordinates in the frequency plane, and the inverse 2D Fourier transform can then be applied to recover μ.

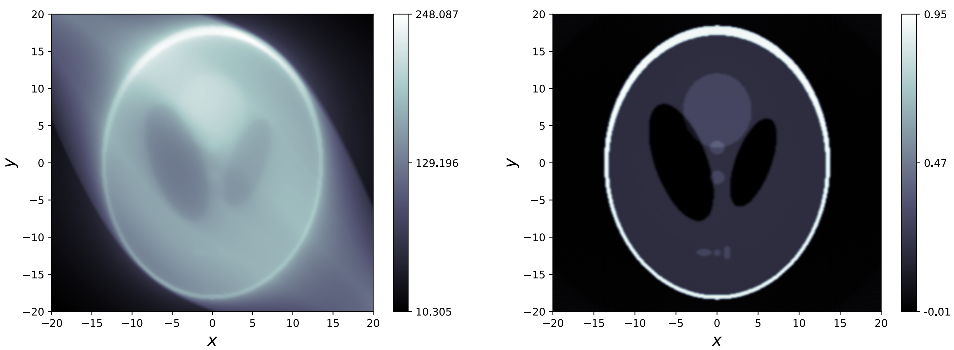
In three dimensions, the situation depends on the acquisition geometry. For parallel-beam CT, an analogue of the central slice theorem still applies:

the 2D Fourier transform of each projection corresponds to a plane through the origin of the object's 3D Fourier transform, and exact inversion is possible with sufficient angular coverage. In cone-beam geometry, however, this direct correspondence is lost. The 2D Fourier transform of a projection does not map to a single plane but to curved manifolds in Fourier space, and exact inversion cannot be obtained by Fourier methods. This motivates the reconstruction approaches discussed in the next section.

2.2 Filtered Backprojection

Filtered Backprojection (FBP) is the most widely used analytical method for reconstructing an image from projection data. As the name suggests, it consists of two main steps: filtering of the projections, followed by backprojection onto the image domain. The method is based on the observation that the backprojection operator smears projection values across the image domain and produces a blurred reconstruction. To counteract this, a high-pass filter is applied before the backprojection to enhance the data's high-frequency components, as it is shown in Figure 2.3.

Let f denote the unknown function representing the object. In the 2D parallel-



(a) Backprojection

(b) Filtered Backprojection

Figure 2.3: Comparison of 2D reconstruction using backprojection and filtered backprojection.

beam setting, the projection data is $g = \mathcal{R}f$, where $\mathcal{R}$ is the Radon transform. Let v be the 1D ramp filter, defined in the Fourier domain as $\hat{v}(\omega) = |\omega|$, and let $\mathcal{R}^*$ denote the backprojection operator (the adjoint of $\mathcal{R}$). Then the FBP reconstruction takes the form:

$$f = \mathcal{R}^*(v * g) = \mathcal{R}^*\mathcal{F}^{-1}(\hat{v}(\omega) \cdot \mathcal{F}g), \tag{2.3}$$

where $*$ denotes the convolution in the detector coordinate. The filtering step can be implemented in the frequency domain by multiplying $\mathcal{F}g$ with $\hat{v}$ and then applying the inverse Fourier transform. The backprojection is then applied to the filtered projections to obtain the reconstructed image. In practice, the filter $\hat{v}$ is often regularized by multiplying with a window function to reduce sensitivity to noise, leading to variants such as the Ram-Lak filter, Shepp-Logan, and cosine filters [20].

In the 3D setting, the standard acquisition geometry is cone-beam, where X-rays diverge from a single focal point and illuminate a two-dimensional detector. In this case, FBP does not yield an exact reconstruction as it does in the 2D parallel-beam case. The reason is that a cone-beam projection does not correspond to a single plane in the object's Fourier domain, so the Fourier Slice Theorem cannot be applied directly. As a result, only approximate FBP-type formulas are possible. The most widely used method is the Feldkamp–Davis–Kress (FDK) algorithm, which can be viewed as an approximate extension of FBP to circular cone-beam trajectories. A schematic expression for FDK can be written as:

$$f \simeq \mathcal{R}^*(v * Wg) = \mathcal{R}^*(\mathcal{F}^{-1}(\hat{v}(\omega) \cdot \mathcal{F}(Wg))) \tag{2.4}$$

where v is the ramp filter applied along detector rows, W is a geometry-dependent weight factor, and $\mathcal{R}^*$ is the cone-beam backprojection [21].

Although approximate, the algorithm was shown to yield accurate reconstructions for sufficiently small fan angles. [1, 21] In the absence of an analytically exact inversion formula for cone-beam geometries, its satisfactory performance under this condition, together with its simple and intuitive FBP-like structure, established it as the standard approach for three-dimensional cone-beam CT reconstruction.

2.3 Katsevich Inversion Formula

An important breakthrough happened in the early 2000s, when Alexander Katsevich introduced the first exact analytical reconstruction formula for helical cone-beam CT. Unlike previous methods, the Katsevich formula is derived directly from the geometry of the acquisition and satisfies the condition for exactness in the continuous setting, known as the Tuy's completeness condition [22].

What makes the Katsevich approach particularly remarkable is that, while grounded in rigorous mathematical theory, it preserves the familiar structure of filtered backprojection. In this sense, it not only closes a theoretical gap but also offers a practical and interpretable framework for exact 3D reconstruction. The derivation of the formula is built on the following key result:

Theorem 1 (Shift-Invariant FBP Reconstruction)**.** *Let Γ be a 1-smooth curve in $\mathbb{R}^3$ with parametrization $y(s)$, $s \in \mathbb{R}$, and let $x \in \mathbb{R}^3 \setminus \Gamma$. Suppose that a bounded function $w(x, y)$ and a smooth unit vector field $V(x, y)$ orthogonal to*

$$\beta(s, x) = \frac{x - y(s)}{\|x - y(s)\|}$$

are chosen for $y \in \Gamma$, with $w(x, y) \neq 0$. For an arbitrary function f with compact support in $\mathbb{R}^3$, we have

$$f(x) = -\frac{1}{2\pi^2} \int_\Gamma \frac{w(x, y(s))}{\|x - y(s)\|} \int_0^{2\pi} \frac{\partial}{\partial q} g((y(q), \theta(s, x, \gamma))\bigg|_{q=s} \frac{d\gamma}{\sin\gamma}\, ds, \tag{2.5}$$

where

$$\theta(s, x, \gamma) = \cos\gamma\, \beta(x, y(s)) + \sin\gamma\, V(x, y(s)) \quad \gamma \in [0, 2\pi),$$

assuming that

$$\sum_{y' \in \Gamma, \langle x - y', \xi \rangle = 0} w(x, y')\ \mathrm{sgn}(\langle \xi, V(x, y') \rangle)\, \mathrm{sgn}(\langle \xi, y' \rangle) = 1 \tag{2.6}$$

holds for almost any $\xi \in S^2$.

Proof. Let H_y be the plane spanned by β and V. The vector $h(t, \gamma) = aV + b\beta$ runs through the plane H_y where $a = t\sin\gamma$ and $b = t\cos\gamma$. Recalling the cone-beam transform definition $g(y, \theta(s, x, \gamma)) = \int_0^\infty f(y + h(t, \gamma))dt$ and operating the change of variables from polar to cartesian coordinates $dh =$

$t\,dt\,d\gamma$, we get:

$$
\begin{aligned}
&\int_0^{2\pi} g(y,\theta(s,x,\gamma))\frac{d\gamma}{\sin\gamma} \\
&= \int_0^{2\pi}\int_0^{\infty} f(y+h(t,\gamma))\frac{d\gamma\,dt}{\sin\gamma} \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(y+h)\frac{dh}{t\sin\gamma} \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(y+h)\frac{da\,db}{a} \\
&= \int_{H_y} f(y+h)\frac{da\,db}{a}.
\end{aligned}
$$

Now taking the inner integral of Equation (2.5), we can fill in the result just obtained to get:

$$
\begin{aligned}
&\frac{1}{\|x-y(s)\|}\int_0^{2\pi}\frac{\partial}{\partial q} g((y(q),\theta(x,y,\gamma))\Big|_{q=s}\frac{d\gamma}{\sin\gamma} \qquad (2.7)\\
=&\frac{1}{\|x-y(s)\|}\frac{\partial}{\partial q}\left(\int_0^{2\pi} g((y(q),\theta(x,y,\gamma))\frac{d\gamma}{\sin\gamma}\right)\Big|_{q=s} \\
=&\frac{1}{\|x-y(s)\|}\left(\frac{\partial}{\partial q}\int_{H_y} f(y(q)+h)\frac{da\,db}{a}\right)\Big|_{q=s} \\
=&\frac{1}{\|x-y(s)\|}\int_{H_y}\frac{\partial}{\partial q} f(y(q)+h)\Big|_{q=s}\frac{da\,db}{a}
\end{aligned}
$$

Expressing f in Fourier domain and changing the order of integration:

$$
=\frac{1}{\|x-y(s)\|}\int_{H_y}\frac{\partial}{\partial q}\int_{\mathbb{R}^3}\hat{f}(\xi)e^{i\langle\xi,y(q)+h\rangle}\Big|_{q=s}\frac{da\,db}{a}\,d\xi \tag{2.8}
$$

$$
=\frac{1}{\|x-y(s)\|}\int_{\mathbb{R}^3}\hat{f}(\xi)\int_{H_y}\frac{\partial}{\partial q}e^{i\langle\xi,y(q)+h\rangle}\Big|_{q=s}\frac{da\,db}{a}\,d\xi \tag{2.9}
$$

$$
=\frac{i}{\|x-y(s)\|}\int_{\mathbb{R}^3}\hat{f}(\xi)\int_{H_y} e^{i\langle\xi,y+h\rangle}\langle\xi,y'\rangle\frac{da\,db}{a}\,d\xi \tag{2.10}
$$

$$
=\frac{i}{\|x-y(s)\|}\int_{\mathbb{R}^3}\hat{f}(\xi)e^{i\langle\xi,y\rangle}\langle\xi,y'\rangle\int_{H_y} e^{ia\langle\xi,V\rangle}e^{ib\langle\xi,\beta\rangle}\frac{da\,db}{a}\,d\xi, \tag{2.11}
$$

using the definition of h in the last equality. Note that :

$$\int_{-\infty}^{\infty} \frac{e^{ia\langle \xi, V\rangle}}{a}\, da = i\pi \operatorname{sgn}(\langle \xi, V\rangle), \tag{2.12}$$

$$\int_{-\infty}^{\infty} e^{ib\langle \xi, \beta\rangle}\, db = 2\pi\delta(\langle \xi, \beta\rangle) \tag{2.13}$$

Therefore 2.11 becomes:

$$= \frac{-2\pi^2}{\|x - y(s)\|} \int_{\mathbb{R}^3} \hat{f}(\xi) e^{i\langle \xi, y\rangle} \langle \xi, y'\rangle \operatorname{sgn}(\langle \xi, V\rangle)\delta(\langle \xi, \beta\rangle)\, d\xi \tag{2.14}$$

$$= -2\pi^2 \int_{\mathbb{R}^3} \hat{f}(\xi) e^{i\langle \xi, y\rangle} \langle \xi, y'\rangle \operatorname{sgn}(\langle \xi, V\rangle)\delta(\langle \xi, x - y\rangle)\, d\xi \tag{2.15}$$

where we used the definition of β and the property of the dirac distribution δ for which $\frac{1}{a}\delta(x) = \delta(ax)$, $a > 0$ to take the norm out of δ. Finally, using the property $F(c)\delta(c) = F(0)\delta(c)$ for all suitable functions F with $c = x - y$ and $F(c) = e^{i\langle \xi, x-c\rangle} = e^{i\langle \xi, y\rangle}$, we obtain:

$$= -2\pi^2 \int_{\mathbb{R}^3} \hat{f}(\xi) e^{i\langle \xi, x\rangle} \langle \xi, y'\rangle \operatorname{sgn}(\langle \xi, V\rangle)\delta(\langle \xi, x - y\rangle)\, d\xi. \tag{2.16}$$

Now consider the whole right-hand side of Equation (2.5). Filling in it the result from 2.16 and changing the order of the integrals we can write it as:

$$\int_{\mathbb{R}^3} \hat{f}(\xi) e^{i\langle \xi, x\rangle} K(x, \xi)\, d\xi, \tag{2.17}$$

where

$$K(x, \xi) = \int_{\Gamma} w(x, y) \langle \xi, y'\rangle \operatorname{sgn}(\langle \xi, V\rangle)\delta(\langle \xi, x - y\rangle)\, ds \tag{2.18}$$

$$= \int_{\Gamma} w(x, y) \operatorname{sgn}(\langle \xi, V\rangle)\delta(\eta) \frac{\langle \xi, y'\rangle}{|\langle \xi, y'\rangle|}\, d\eta \tag{2.19}$$

by changing variable from s to $\eta = \langle \xi, x - y(s)\rangle$, so $d\eta = |\langle \xi, y'\rangle| ds$. This reduces to the sum:

$$\sum_{y' \in \Gamma, \langle x - y', \xi\rangle = 0} w(x, y')\ \operatorname{sgn}(\langle \xi, V(x, y')\rangle) \operatorname{sgn}(\langle \xi, y'\rangle). \tag{2.20}$$

Equation (2.20) is equal to 1 by assumption. Therefore we conclude that

$K(x,\xi) \equiv 1$ and Equation (2.17) becomes:

$$\int_{\mathbb{R}^3} \hat{f}(\xi) e^{i\langle \xi, x\rangle}\, d\xi, \tag{2.21}$$

which is by definition of the Fourier transform equal to $f(x)$. □

The challenge of this result lies in finding and proving for which choice of $w(s,x)$ and $V(s,x)$ the equality 2.6 is correct. Below we consider the case when the curve is a helix and present the results obtained by Katsevich in [7].

2.3.1 Helical Geometry

Before presenting the formula, we need to introduce the geometry configuration and explain some related concepts. The geometry of the scan comprises two main elements: the X-ray source and the detector. The source and the detector move around the patient in a specular way with respect to the rotation axis, in our case the z-axis, as illustrated in Figure 2.4.

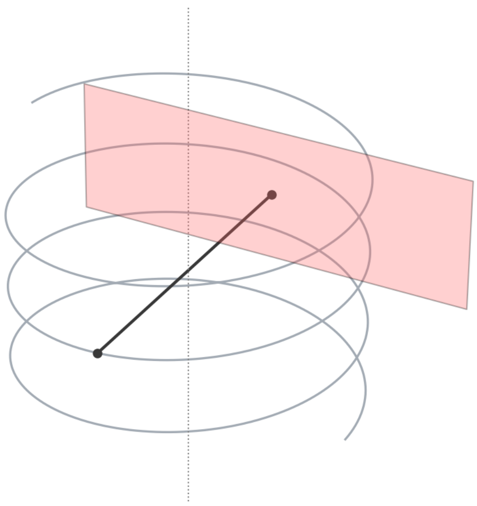


Figure 2.4: Helical geometry configuration.

We describe the configuration by introducing a local reference frame at each rotation angle s. This frame is defined by three vectors:

- $\mathbf{e}_w$: the constant unit vector aligned with the z-axis.

- $\mathbf{e}_v(s)$: the unit vector pointing from the source toward the center of the detector;
- $\mathbf{e}_u(s)$: the cross product of $\mathbf{e}_v(s)$ and $\mathbf{e}_w$;

The source follows a helical trajectory. Using the reference frame defined by $\mathbf{e}_u(s)$, $\mathbf{e}_v(s)$, $\mathbf{e}_w$, the helical curve can be expressed as:

$$y(s) = -R\,\mathbf{e}_v(s) + p(s)\,\mathbf{e}_w, \tag{2.22}$$

where R is the radial distance of the source from the z-axis, and $p(s)$ is the axial displacement along the rotation axis. By convention, most of the literature adopts the following specific basis:

$$\mathbf{e}_u(s) = [-\sin s,\ \cos s,\ 0]^T, \quad \mathbf{e}_v(s) = [-\cos s,\ -\sin s,\ 0]^T, \tag{2.23}$$

In this case, the initial configuration begins at:

$$\mathbf{e}_u(0) = \mathbf{e}_y, \quad \mathbf{e}_v(0) = -\mathbf{e}_x, \quad \mathbf{e}_w = e_z,$$

and rotates counter-clockwise in the xy-plane as the angle s increases. With this choice, the source trajectory becomes:

$$y(s) = [R\cos s,\ R\sin s,\ p(s)]^T, \tag{2.24}$$

i.e., a helix with radius R and general three-time differentiable function $p(s)$. A common special case considered is when $p(s) = \frac{Ps}{2\pi}$, where the pitch P is constant.

Once the helical source trajectory has been defined, we focus on the problem of reconstructing the object at a specific point x in the reconstruction volume. A naive approach would be to use all cone-beam projections acquired along the entire helical path. However, this would involve a significant amount of redundant data which cause artifacts in the reconstructed image. An exact reconstruction of $f(x)$ in practice only requires a subset of the acquired projections. This minimal subset can be identified using the concept of $PI-$lines.

A $PI-$line (see Figure 2.5) is a straight line that connects two source positions $y(s_1)$ and $y(s_2)$ that belong to the same helical turn, meaning their

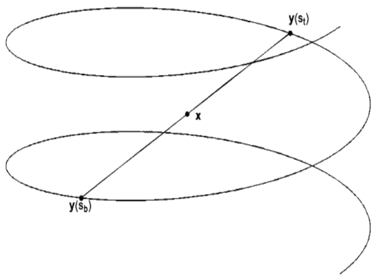


Figure 2.5: $PI-$line of x.

rotation angles satisfy:

$$|s_2 - s_1| < 2\pi.$$

A remarkable property of helical geometries is that every point x within the helix volume is intersected by exactly one such $PI-$line, provided that $p'(s) + p'''(s) > 0 \quad a.e.$ [8, 23]. Denoting the endpoints of the $PI-$line corresponding to the volume point x as $s_b(x), s_t(x)$, we define the $PI-$interval as:

$$I_{\mathrm{PI}}(x) = [s_b(x), s_t(x)].$$

It has been shown that integrating the cone-beam data over this interval, using only those projections where the source lies within I_{PI}, is sufficient to reconstruct $f(x)$ exactly; all others are irrelevant for that point [24]. To impose these bounds to the integration, we therefore set $w(s, x) = \mathbb{1}_{I_{\mathrm{PI}}}$.

Note that this approach has the significant limitation that the relevant boundaries of the $PI-$interval depend on the specific point x to be reconstructed. Therefore, the integration bounds would vary spatially across the volume, making it complex to have a modular algorithm that can be applied to the whole volume at once. So in practice, we use a strategy proposed by Tam and Danielsson that allows to obtain a perfect reconstruction integrating over the full set of projections, by restricting the data on the detector, using the so-called Tam-Danielsson (TD) window [25]. This concept will be further explained in Section 3.1.1.

Going further, to complete the formulation we not only need to restrict the amount of projection data used, but also need a precise way to organize and process this data. In particular, we must decide how to filter the measured cone-beam data before backprojecting it into the reconstruction volume. For this we introduce $\kappa-$planes and $\kappa-$lines, which provide the path along which the projection data has to be filtered. We define a $\kappa-$plane $\kappa(s_0, \psi)$ as the two-dimensional plane that intersects the helical source trajectory at three points

$$y(s_0), \quad y(s_0+\psi), \quad y(s_0+2\psi),$$

for some $\psi \in (-\pi, \pi)$. The parameter ψ controls the spacing of these intersection points on the helix, and hence determines the plane. The intersection of $\kappa-$planes $\kappa(s,\psi)$ and the detector plane $DP(s)$ gives the $\kappa-$lines. These $\kappa-$lines play a critical role in Katsevich's algorithm: the cone-beam data are first filtered along these lines, and only afterward backprojected along rays to reconstruct the object. From this definition, the unit normal vector to the $\kappa-$plane $\kappa(s_0, \psi)$ can be expressed as:

$$n(s,\psi) = \frac{(y(s+\psi) - y(s)) \times (y(s+2\psi) - y(s))}{||(y(s+\psi) - y(s)) \times (y(s+2\psi) - y(s))||}\mathrm{sgn}(\psi), \quad \psi \in (-\pi, \pi) \tag{2.25}$$

For a given direction u and parameter value s, define $m(s,u)$ as the normal vector $n(s,\psi)$ corresponding to the smallest $|\psi|$ for which the associated $\kappa-$plane $\kappa(s,\psi)$ contains u in its span:

$$m(s,u) = n(s,\psi^*), \quad \psi^* = \arg\min_{|\psi|}\{u \in \mathrm{span}(\kappa(s,\psi))\}.$$

Let β be as defined in Equation (2.5) (*i.e.*, the direction from the source point $y(s)$ to a point x in the reconstruction domain). Define H_y as the $\kappa-$plane orthogonal to $m(s,\beta(s,x))$, *i.e.*, the plane determined by β and its orthogonal complement in the $\kappa-$plane. The second vector spanning this $\kappa-$plane is defined by:

$$V(s,x) = \beta(s,x) \times m(s,\beta). \tag{2.26}$$

The pair $(\beta(s,x), V(s,x))$ forms an orthonormal basis for the the $\kappa-$plane H_y, which is used to define the direction of filtering in the reconstruction formula.

It is important to emphasize that choices of these geometric elements $w(s,x)$, $V(s,x)$ and H_y are not arbitrary. The use of the $PI-$line interval to restrict the range of integration, the construction of the vector field $V(s,x)$, and the orientation of the $\kappa-$plane H_y are all determined by the conditions under which the inversion formula of Theorem 1 is valid. With this geometrical elements in place, we are now ready to state the exact inversion formula for helical cone-beam geometries, first proposed by Katsevich in [7, 8].

Theorem 2 (Katsevich's Inversion Formula)**.** *Set $w(s,x)$ equal to the indicator function for $I_{PI}(x)$, the vector field $V(s,x) = \beta(s,x) \times m(s,\beta)$ (β and m as defined above), and the trajectory*

$$y(s) = [R\cos(s),\, R\sin(s),\, p(s)]^T,$$

with

$$p'(s) + p'''(s) > 0 \quad a.e.;$$

Then the assumption in Equation (2.6) holds and Equation (2.5) becomes:

$$f(x) = -\frac{1}{2\pi^2}\int_{I_{\mathrm{PI}}(x)} \frac{1}{\|x-y(s)\|}\int_0^{2\pi} \frac{\partial}{\partial q} Df(y(q),\theta(s,x,\gamma))\Big|_{q=s} \frac{d\gamma}{\sin\gamma}\, ds \tag{2.27}$$

Proof. It is trivial to see that, given that Equation (2.6) holds, Equation (2.5) simplifies into Equation (2.27). To show that indeed for $w(s,x) = \mathbb{1}_{I_{\mathrm{PI}}(x)}$ and $V(s,x) = \beta(s,x) \times m(s,\beta)$, Equation (2.6) holds, refer to [7]. □

2.3.2 Operator-Based Formulation

To see that the Katsevich's formula is of the FBP-type, we can express it in an alternative form. Define:

$$g'(s,\eta(s,x)) = \frac{\partial}{\partial q} Dg(y(q),\eta(s,x))\Big|_{q=s} \tag{2.28}$$

as the derivative of the data function for an arbitrary direction in the $\kappa-$plane $\eta(s,x) = \cos\delta\beta(s,x) + \sin\delta V(s,x), \quad \delta \in [0,2\pi)$. Let:

$$g^{\mathrm{F}}(s,\eta(s,x)) := \int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma)) g'(s,\cos(\delta-\gamma)\beta(s,x) - \sin(\delta-\gamma)V(s,x))\, d\gamma.$$

Then, considering the direction of β in the $\kappa-$plane,

$$g^{\mathrm{F}}(s,\beta(s,x)) = \int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma))g'(s,\cos(-\gamma)\beta(s,x) - \sin(-\gamma)V(s,x))\,d\gamma \tag{2.29}$$

$$= \int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma))g'(s,\cos(\gamma)\beta(s,x) + \sin(\gamma)V(s,x))\,d\gamma \tag{2.30}$$

$$= \int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma))g'(s,\theta(s,x,\gamma))\,d\gamma \tag{2.31}$$

$$= \frac{1}{\pi}\int_0^{2\pi} \frac{\partial}{\partial q} Df(y(q),\theta(s,x,\gamma))\Big|_{q=s} \frac{1}{\sin\gamma}\,d\gamma \tag{2.32}$$

where the last step comes from the fact that $k_H(\sin(\gamma)) = \frac{1}{\pi\sin(\gamma)}$, and thus to summarize:

$$\frac{1}{\pi}\int_0^{2\pi} \frac{\partial}{\partial q} Df(y(q),\theta(s,x,\gamma))\Big|_{q=s} \frac{1}{\sin\gamma}\,d\gamma = g^F(s,\beta(s,x))$$

Thanks to this equality, we can express Equation (2.27) in the new form:

$$f(x) = -\frac{1}{2\pi}\int_{I_{\mathrm{PI}}(x)} \frac{1}{\|x-y(s)\|} g^{\mathrm{F}}(s,\beta(s,x))\,ds \tag{2.33}$$

$$= -\frac{1}{2\pi}\int_{I_{\mathrm{PI}}(x)}\int_0^{2\pi} \frac{k_{\mathrm{H}}(\sin(\gamma))g'(s,\theta(s,x,\gamma))}{\|x-y(s)\|}\,d\gamma\,ds \tag{2.34}$$

From this last formulation, we see that the Katsevich formula involves taking the derivative of the projection data (g'), performing a 1D convolution between g' and k_{H} $\left(\int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma))g'(s,\theta(s,x,\gamma))\,d\gamma \text{ in Equation (2.31)}\right)$, and integrating the result over a bounded interval $\left(-\frac{1}{2\pi}\int_{I_{\mathrm{PI}}(x)} \dots ds\right)$. This highly resembles the composition of a convolution operation and a adjoint operator that we find in the FBP formulation in Equation (2.3). However, Katsevich's method does not directly apply a Fourier-domain filter.

Instead, the formula is decomposed in:

$$f(x) = \mathcal{R}_w^* \mathcal{D}(g), \tag{2.35}$$

where:

- $\mathcal{R}_w^*$ is the backprojection operator for the cone beam (the w stands for weighted, as we will see later in Chapter 3),

- $\mathcal{D}$ is a differentiation-based filter applied to projection data g,

where $\mathcal{D}$ involves convolving the derivative of the data. This operator-based formulation will be useful in the implementation of the algorithm in local detector coordinates discussed in Section 3.1.

Since the projection data g are acquired on the detector, they depend on the specific choice of detector and local sampling system. Therefore in the next sections we outline the different detector configurations and the expression of the Katsevich inversion formula for a flat detector in Section 2.4 and for a curved detector in Section 2.5.

2.4 Flat Detector

The detector plane lies in the $(\mathbf{e}_u(s), \mathbf{e}_w)$ plane and is centered at:

$$DP(s) = y(s) + D\,\mathbf{e}_v(s).$$

Let (u, w) be the coordinates on the detector plane, where u and w are

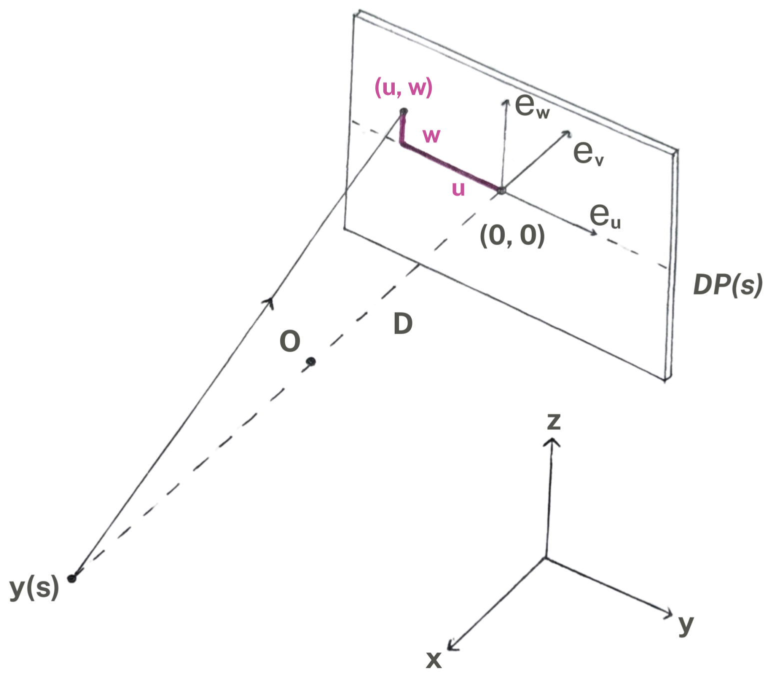


Figure 2.6: Flat detector geometry.

respectively the horizontal and the vertical displacement from the detector center. Any point on the detector is given in global coordinates by:

$$d(s,u,w) = y(s) + D\,\mathbf{e}_v(s) + u\,\mathbf{e}_u(s) + w\,\mathbf{e}_w. \tag{2.36}$$

The direction of X-rays hitting the detector at point (u,w) is:

$$\theta_f(s,u,w) = \frac{1}{\sqrt{u^2 + D^2 + w^2}}\left(u\,\mathbf{e}_u(s) + D\,\mathbf{e}_v(s) + w\,\mathbf{e}_w\right). \tag{2.37}$$

Conversely, given a ray direction θ from $y(s)$, the detector coordinates (u,w) where it intersects the plane are given by:

$$u = D\frac{\langle \theta, \mathbf{e}_u(s)\rangle}{\langle \theta, \mathbf{e}_v(s)\rangle}, \quad w = D\frac{\langle \theta, \mathbf{e}_w\rangle}{\langle \theta, \mathbf{e}_v(s)\rangle}. \tag{2.38}$$

2.4.1 Katsevich in Local Flat Detector Coordinates

We reformulate the Katsevich backprojection formula using the coordinate system of the flat detector. The derivation follows the approach presented in [16]. For brevity, only the main results are summarized here; for a detailed step-by-step explanation, the reader is referred to the original work of Wunderlich.

We begin by rewriting the derivative term 2.28 in local coordinates:

$$\begin{aligned} g_1(s,u,w) &= \frac{\partial}{\partial q} g(q,u,w)\Big|_{q=s} \\ &= \left(\frac{\partial g}{\partial q} + \frac{u^2 + D^2}{D}\frac{\partial g}{\partial u} + \frac{uw}{D}\frac{\partial g}{\partial w}\right)\Big|_{q=s} \end{aligned} \tag{2.39}$$

Equation (2.39) shows how the derivative of the data function with respect to s can be expanded in terms of local detector variables (u,w).

The next step is to identify the equations of the $\kappa-$lines, the lines that correspond to the intersection of X-rays lying on the $\kappa-$plane with the detector plane. The $\kappa-$plane is defined by the source position $y(s)$ and the unit vector $n(s,\psi)$ (see Equation (2.25)), with ψ being the angular offset defining the PI-segment $[s, s + 2\psi]$. Following Katsevich's formulation [8], the $\kappa-$lines are computed numerically solving:

$$(x - y(s)) \cdot n(s,\psi) = 0, \quad s \in I_{\mathrm{PI}}(x), \tag{2.40}$$

for every source position s; that is, we find the $\kappa-$lines by finding points on the detector where the vector from $y(s)$ to the detector point is orthogonal to the normal vector of the $\kappa-$plane.

Let $x = \mathbf{d}(u, w)$ be a point on the detector. $\mathbf{d}$ lies on the $\kappa-$plane that passes through the point $y(s)$ and has normal vector $n(s, \psi)$ if the vector from $y(s)$ to $\mathbf{d}$ is orthogonal to $n(s, s_2)$, that is:

$$\begin{aligned} &(\mathbf{d}(u,w) - y(s)) \cdot n(s,\psi) = 0 \\ \implies &(u\mathbf{e}_u(s) + D\mathbf{e}_v(s) + w\mathbf{e}_w(s)) \cdot n(s,\psi) = 0, \end{aligned}$$

using Equation (2.37) since $\mathbf{d}$ lies on the detector plane $DP(s)$. This equation is linear in w so we can rearrange it and conclude that, for a general variable pitch $p(s) \in C^3(\mathbb{R})$, the equation for the $\kappa-$line of angle ψ for the view point s is:

$$w_\kappa(u) = -\frac{D\mathbf{e}_v(s) \cdot n(s,\psi) + u\mathbf{e}_u(s) \cdot n(s,\psi)}{\mathbf{e}_w \cdot n(s,\psi)}. \tag{2.41}$$

In the specific case where the pitch is constant, *i.e.*, $p(s) = \frac{hs}{2\pi}$, the equation

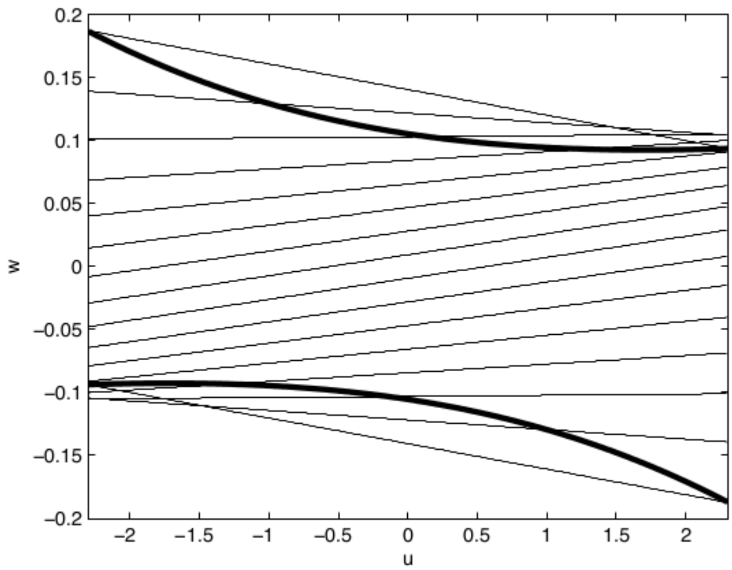


Figure 2.7: $\kappa-$lines on flat detector for different ψ and TD Window (in bold).

for the $\kappa-$line of angle ψ becomes constant over s:

$$w_\kappa(u) = \frac{Dh}{R}\left(\psi + \frac{\psi}{D\tan\psi}u\right). \tag{2.42}$$

Figure 2.7 illustrates the $\kappa-$lines for the constant pitch case. Moreover, the derivation of this special case can be verified in Appendix A.

The next step is to express $g^F(s,\beta)$ (the filtered projection data) in local coordinates. To do this, we consider two rays on the $\kappa-$plane intersecting the detector: one in the direction β and another in a perturbed direction $\theta(s,x,\gamma) = \cos\gamma\beta(s,x) + \sin\gamma V(s,x)$ parameterized by angle γ. Let $(u, w_\kappa(u))$ and $(u', w'_\kappa(u'))$ denote the intersection points of these rays with the detector and denote the respective distances from the source to these detector points as r and r'. Then we can relate g^F to the locally defined filtered data g_1 in the following way:

$$g^F(s,\beta) = \int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma))g'(s,\theta(s,x,\gamma))\, d\gamma \tag{2.43}$$

simplifies into:

$$g^F(s,\beta) = -\frac{r}{D}\int_{-\infty}^{\infty} k_H(u-u')\frac{D}{r'}g_1(s,u',w'_\kappa(u'))du', \tag{2.44}$$

With this, the integrand in Equation (2.34) can be rewritten using local coordinates:

$$\frac{-1}{|x-y(s)\|}g^F(s,\beta) = \frac{1}{R+\langle x,\mathbf{e}_v\rangle}g^F(s,u,w_\kappa(u)), \tag{2.45}$$

where

$$\begin{aligned} g^F(s,u,w_\kappa(u)) &:= \int_{-\infty}^{\infty} k_H(u-u')\frac{Dg_1(s,u',w'_\kappa(u'))}{\sqrt{(u')^2+D^2+(w'_\kappa(u'))^2}}du' \\ &= -\frac{D}{r}g^F(s,\beta). \end{aligned} \tag{2.46}$$

Collecting all the previous results allows us to express the Katsevich's inversion formula in flat detector coordinates as:

$$
\begin{aligned}
f(x) &= \frac{1}{2\pi}\int_{I_{\mathrm{PI}}(x)} \frac{g^F(s,u^*(s,x),w^*(s,x))}{R+\langle x,\mathbf{e}_v\rangle}\,ds \\
g^F(s,u^*(s,x),w^*(s,x)) &= \int_{-\infty}^{\infty} D\frac{k_H(u^*-u')g_1(s,u',w'_\kappa(u'))}{\sqrt{(u')^2+D^2+(w'_\kappa(u'))^2}}\,du' \\
g_1(s,u',w'_\kappa(u')) &= \left(\frac{\partial g}{\partial q}+\frac{(u')^2+D^2}{D}\frac{\partial g}{\partial u'}+\frac{u'w'_\kappa(u')}{D}\frac{\partial g}{\partial w'_\kappa}\right)\Big|_{q=s}
\end{aligned}
$$

with

$$
\begin{aligned}
u^*(s,x) &= D\frac{\langle\beta,\mathbf{e}_u(s)\rangle}{\langle\beta,\mathbf{e}_v(s)\rangle} = \frac{D\langle x,\mathbf{e}_u(s)\rangle}{R+\langle x,\mathbf{e}_v(s)\rangle}, \\
w^*(s,x) &= D\frac{\langle\beta,\mathbf{e}_w\rangle}{\langle\beta,\mathbf{e}_v(s)\rangle} = D\frac{(x_3-p(s))}{R+\langle x,\mathbf{e}_v(s)\rangle},
\end{aligned}
$$

where $(u^*(s,x),w^*(s,x))$ is the intersection point of the detector and the ray passing through $y(s)$ and x.

2.5 Curved Detector

The curved detector lies on a cylinder centered at the source $y(s)$ with radius D and spans the $(\mathbf{e}_u(s),\mathbf{e}_w)$ directions. At each rotation angle s, the detector is centered at:

$$DP(s) = y(s) + D\,\mathbf{e}_v(s),$$

Detector coordinates are denoted by (α,w), where:

- α is the angular displacement from the detector center in the $(\mathbf{e}_u(s),\mathbf{e}_v)$ plane (*i.e.*, the arc angle along the cylindrical detector), measured in radians,
- w is the vertical displacement along $\mathbf{e}_w$ from the detector center.

Any point on the curved detector is given in global coordinates by:

$$d(s,\alpha,w) = y(s) + D\left(\cos\alpha\,\mathbf{e}_v(s)+\sin\alpha\,\mathbf{e}_u(s)\right)+w\,\mathbf{e}_w. \tag{2.47}$$

The direction of X-rays hitting the detector at coordinates (α,w) is:

$$\theta_c(s,\alpha,w) = \frac{1}{\sqrt{D^2+w^2}}\left(D\cos\alpha\,\mathbf{e}_v(s)+D\sin\alpha\,\mathbf{e}_u(s)+w\,\mathbf{e}_w\right). \tag{2.48}$$

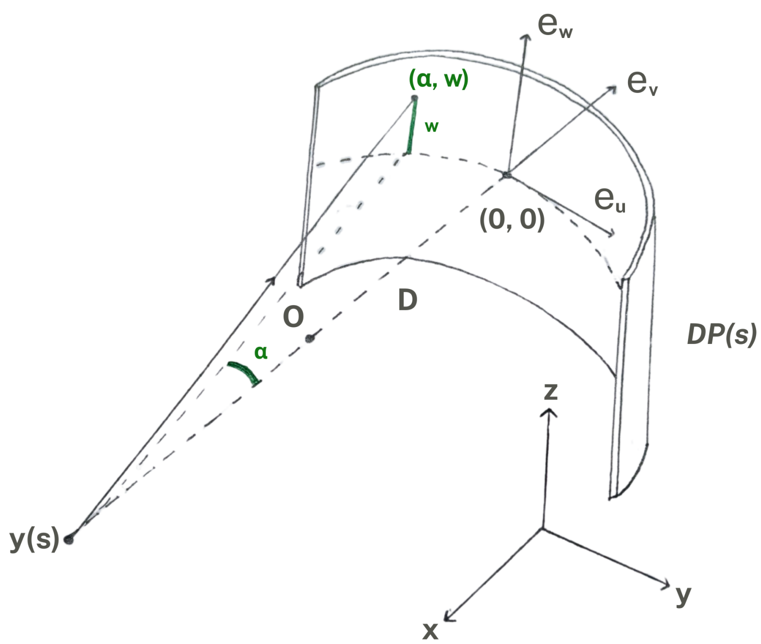


Figure 2.8: Curved detector geometry.

Conversely, given a ray direction θ from $y(s)$, the detector coordinates (α, w) at which it intersects the cylindrical detector are:

$$\alpha = \arctan\left(\frac{\langle\theta, \mathbf{e}_u(s)\rangle}{\langle\theta, \mathbf{e}_v(s)\rangle}\right), \quad w = D\frac{\langle\theta, \mathbf{e}_w\rangle}{\sqrt{\langle\theta, \mathbf{e}_v(s)\rangle^2 + \langle\theta, \mathbf{e}_u(s)\rangle^2}}. \tag{2.49}$$

Note that if, given a X-ray direction θ and the point it hits on the curved detector (α, w_c), one would be interested in knowing the point (u, w_f) that would be hit on a flat detector, this would be given by:

$$u = D\tan\alpha, \quad w_f = \frac{w_c}{\cos\alpha}, \tag{2.50}$$

see the illustration in Figure 2.9. The first equation is trivial to derive. The second equation is derived by equalizing θ_c and θ_f and taking their magnitude

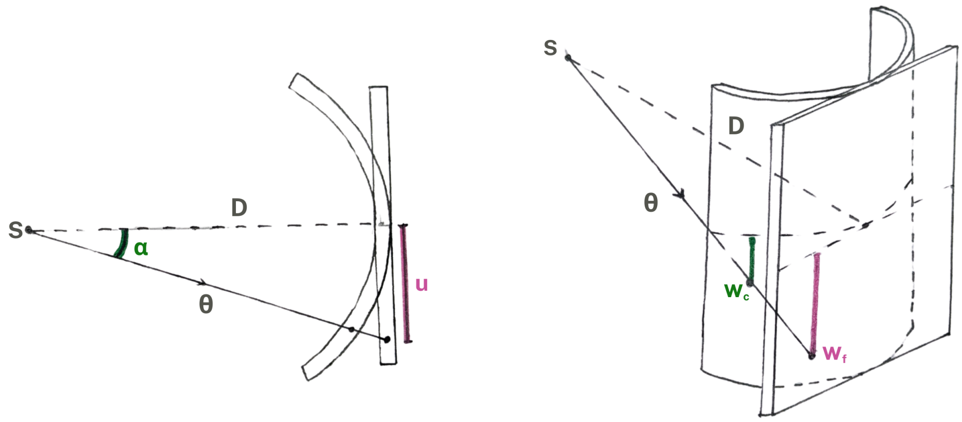


Figure 2.9: Top and side view of a ray θ hitting the curved and the flat detector.

in the $\mathbf{e}_u(s)$ direction:

$$
\begin{aligned}
&\frac{u}{\sqrt{u^2 + D^2 + w_f^2}} = \frac{D \sin \alpha}{\sqrt{D^2 + w_c^2}} \\
\implies &\frac{D \tan \alpha}{\sqrt{D^2 \tan^2 \alpha + D^2 + w_f^2}} = \frac{D \sin \alpha}{\sqrt{D^2 + w_c^2}} \\
\implies &\frac{1}{\cos \alpha \sqrt{D^2 \tan^2 \alpha + D^2 + w_f^2}} = \frac{1}{\sqrt{D^2 + w_c^2}} \\
\implies &\cos^2 \alpha \, (D^2(1 + \tan^2 \alpha) + w_f^2) = D^2 + w_c^2 \\
\implies &D^2 + \cos^2 \alpha \, w_f^2 = D^2 + w_c^2 \\
\implies &w_f^2 = \frac{D^2 + w_c^2 - D^2}{\cos^2 \alpha} \\
\implies &w_f = \frac{w_c}{\cos \alpha}.
\end{aligned}
$$

where we used the formula for u and the identity $1 + \tan^2 \alpha = 1/\cos^2 \alpha$.

2.5.1 Katsevich in Local Curved Detector Coordinates

The derivation proceeds similarly to the flat detector case in Section 2.4.1, with the key difference being the use of a different coordinate system. Specifically, the axial position is parameterized by the angular coordinate α, that accounts for the curvature of the detector. The following steps briefly outline the main changes.

First, the derivative term 2.28 is expressed in local coordinates:

$$\begin{aligned} g_1(s,\alpha,w) &= \frac{\partial}{\partial q} g(q,\alpha,w)\Big|_{q=s} \\ &= \left(\frac{\partial g}{\partial q} + \frac{\partial g}{\partial \alpha}\right)\Big|_{q=s}. \end{aligned} \tag{2.51}$$

Next, we derive the equation of the $\kappa-$lines in curved detector coordinates. A detector point $\mathbf{d}(\alpha, w)$ is expressed in cylindrical form using the local detector basis $\{\mathbf{e}_u(s), \mathbf{e}_w\}$. Substituting $\mathbf{d}$ into the orthogonality condition with the $\kappa-$plane's normal vector $n(s,\psi)$ yields the expression:

$$\begin{aligned} &(\mathbf{d}(\alpha,w) - y(s)) \cdot n(s,\psi) = 0 \\ \implies &(D\sin(\alpha)\mathbf{e}_u(s) + D\cos(\alpha)\mathbf{e}_v(s) + w\mathbf{e}_w(s)) \cdot n(s,\psi) = 0, \end{aligned}$$

Rearranging this equation provides us with the equation for the $\kappa-$line of angle ψ for a general variable pitch $p(s) \in C^3(\mathbb{R})$:

$$w_\kappa(\alpha) = -\frac{D\cos(\alpha)\mathbf{e}_v(s) \cdot n(s,\psi) + D\sin(\alpha)\mathbf{e}_u(s) \cdot n(s,\psi)}{\mathbf{e}_w \cdot n(s,\psi)}. \tag{2.52}$$

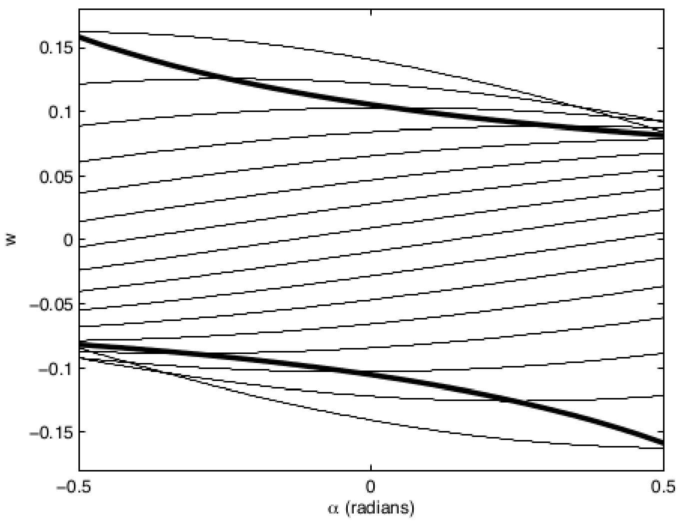


Figure 2.10: $\kappa-$curves on a curved cylindrical detector for various ψ. The bold lines represent the Tam-Danielsson window.

In the case of constant pitch ($p(s) = \frac{hs}{2\pi}$), the equation simplifies to the form:

$$w_\kappa(\alpha) = \frac{Dh}{R}\left(\psi\cos(\alpha) + \frac{\psi}{\tan\psi}\sin(\alpha)\right). \tag{2.53}$$

Figure 2.10 illustrates the $\kappa-$lines for this special case.

Further on, we reformulate $g^F(s,\beta)$ in curved detector coordinates. Here, the result of Equation (2.44) can be extended to the curved case by:

$$g^F(s,\beta) = -\frac{\sqrt{D^2 + w_\kappa(\alpha)^2}}{D}\int_{-\pi/2}^{\pi/2} k_H(\sin(\alpha-\alpha'))\frac{Dg_1(s,\alpha',w'_\kappa(\alpha'))}{\sqrt{D^2+w'_\kappa(\alpha')^2}}d\alpha'. \tag{2.54}$$

Finally, we express the integrand in Equation (2.34) as:

$$\frac{-1}{|x-y(s)\|}g^F(s,\beta) = \frac{1}{R+\langle x,\mathbf{e}_v\rangle}g^F(s,\alpha,w_\kappa(u)), \tag{2.55}$$

where:

$$g^F(s,\alpha,w_\kappa) := \cos\alpha\int_{-\pi/2}^{\pi/2} k_H(\sin(\alpha-\alpha'))\frac{Dg_1(s,\alpha',w'_\kappa)}{\sqrt{D^2+w'_\kappa(\alpha')^2}}d\alpha'.$$

The function g^F is now defined in terms of (α, w_κ) and involves a cosine weight due to the angular detector parameterization.

Putting everything together, Equation (3.1) presents the full Katsevich inversion formula in curved detector coordinates:

$$\begin{aligned}
f(x) &= \frac{1}{2\pi}\int_{I_{\mathrm{PI}}(x)}\frac{g^F(s,\alpha^*(s,x),w^*(s,x))}{R+\langle x,\mathbf{e}_v(s)\rangle}ds,\\
g^F(s,\alpha^*(s,x),w^*(s,x)) &= \cos\alpha\int_{-\pi/2}^{\pi/2}k_H(\sin(\alpha^*(s,x)-\alpha'))\frac{Dg_1(s,\alpha',w'_\kappa(\alpha'))}{\sqrt{D^2+(w'_\kappa(\alpha'))^2}}d\alpha',\\
g_1(s,\alpha',w'_\kappa(\alpha')) &= \left(\frac{\partial g}{\partial q}(q,\alpha',w'_\kappa) + \frac{\partial g}{\partial\alpha}(q,\alpha',w'_\kappa)\right)\Big|_{q=s}.
\end{aligned}$$

with:

$$
\begin{aligned}
\alpha^*(s,x) &= \arctan\left(\frac{\langle \beta, \mathbf{e}_u(s)\rangle}{\langle \beta, \mathbf{e}_v(s)\rangle}\right) = \arctan\left(\frac{\langle x, \mathbf{e}_u(s)\rangle}{R + \langle x, \mathbf{e}_v(s)\rangle}\right), \\
w^*(s,x) &= \cos\alpha^*(s,x) w^*(s,x) = D\frac{\cos\alpha^*(s,x)(x_3 - p(s))}{R + \langle x, \mathbf{e}_v(s)\rangle},
\end{aligned}
$$

where $(\alpha^*(s,x), w^*(s,x))$ is the intersection point of the detector and the ray passing through $y(s)$ and x.

2.6 ODL Library and ASTRA Toolbox

The implementation of the reconstruction methods presented in this thesis is based on the open tools Operator Discretization Library (ODL) [26] and the ASTRA Toolbox [27].

The Operator Discretization Library (ODL) is a Python framework for solving inverse problems, with a particular focus on applications in imaging. One of its defining features is its high modularity: complex pipelines can be built from a small number of well-defined, modular components. Although the operations involved in tomographic reconstruction (such as 3D projections and backprojections) are computationally intensive and often require parallelization, ODL abstracts this complexity by exposing to the user a clean, high-level interface centered on mathematical operators. This enables users to construct and manipulate forward and inverse models, such as the Radon tranform and its adjoint, as algebraic objects, independent of the underlying implementation details.

As its key feature, ODL formulates the inverse problems as operator equations of the form $Af = g$, where A is a forward operator such as the ray transform, f is the unknown object to be reconstructed (typically a function or image), and g is the observed data, for instance the projection data collected by a CT scanner.

While ODL offers a CPU-based backend for basic 2D problems, real-world 3D tomography requires high-performance computations that go beyond what can be achieved with NumPy or SciPy alone. For this reason, ODL is often used in combination with the external backend ASTRA Toolbox.

The ASTRA Toolbox is a high-performance C++ and CUDA library specifically designed for efficient forward and backward projections in 2D and 3D tomography. It includes GPU-accelerated algorithms for cone-beam geometries and supports flexible detector configurations. ODL provides a native interface to ASTRA through its `astra_cuda` backend, making it possible to define an operator symbolically in Python while executing it on the GPU through ASTRA's optimized core.

2.7 Summary

This chapter introduced the principles of tomographic reconstruction and highlighted the limitations of traditional methods like Filtered Backprojection (FBP) when extended to 3D cone-beam CT. While FBP is exact in 2D, its 3D cone-beam counterparts, such as the Feldkamp-Davis-Kress (FDK) algorithm, are only an approximation.

The Katsevich inversion formula provides an analytically exact solution for helical cone-beam CT, relying on a precise geometric understanding of the acquisition. Its exactness is guaranteed by two main elements: restricting the data to the PI-line interval for each reconstruction point, and performing filtering along $\kappa-$lines that adapt to the acquisition geometry. In practice, enforcing the PI-line condition directly is challenging, and the Tam-Danielsson window offers a practical way to limit the detector data to the minimal window that contains all necessary information on the volume for an exact reconstruction.

Despite its geometric complexity, the formula preserves the structure of FBP: a one-dimensional filtering followed by backprojection. This structure allows it to be implemented modularly using the Operator Discretization Library (ODL), which models reconstruction as compositions of mathematical operators and interfaces efficiently with the GPU-accelerated ASTRA toolbox. In the next chapter, we discuss the numerical operations needed to implement the Katsevich formula and we show how it can be integrated into the ODL framework.

Chapter 3

Methods

This chapter moves from theory to methodology. It describes the engineering approach, numerical strategies, and implementation details adopted to realize the Katsevich reconstruction within ODL, using ASTRA as computational backend. The goal is to translate the continuous formulation into a modular and efficient numerical algorithm, in a way that integrates smoothly with the existing ODL framework.

We first present an overview of the algorithm's logical structure, identifying the main operators and their roles in the reconstruction pipeline (Section 3.1). This is followed by a discussion of the practical implementation within ODL (Section 3.2), including the design of custom operators and their composition into the full reconstruction workflow.

3.1 Algorithm Overview

This section will describe the algorithm created to carry out Katsevich's reconstruction based on the formulas derived in Section 2.4.1 for the flat case and Section 2.5.1 for the curved case.

In Equation (2.35) we mentioned that the Katsevich reconstruction formula has the operator-based formulation:

$$f(x) = \mathcal{R}_w^* \mathcal{D}(g), \tag{3.1}$$

This holds also for the Katsevich formula expressed in local detector coordinates: the filtering $\mathcal{D}$ takes care of the map $g \mapsto g^F$ and $\mathcal{R}_w^*$ operates the weighted backprojection $g^F \mapsto f(x)$.

The numerical implementation details of this algorithm have been outlined by Noo et al. in [15]. The filtering operator $\mathcal{D}$ is decomposed into multiple sequential steps, CF1-CF6 for curved detectors and FF1-FF5 for flat detectors, each involving specific numerical operations and discretization strategies. The weighted backprojection $\mathcal{R}_w^*$ with voxel-dependent boundaries is split in two parts: the TD windowing of the filtered projection data and the unbounded backprojection operation.

The sections below explain the numerical details of the filtering steps and of the TD windowing for both curved and flat detector geometries. Note that the backprojection step is delegated to the underlying computational backend (ASTRA in our case) and does not require explicit numerical implementation.

3.1.1 Katsevich Algorithm for Curved Detector

Discretize the detector coordinates via:

$$\begin{aligned} \alpha_i &= (i - (N_\alpha - 1)/2)\Delta\alpha \quad i = 0, ..., N_\alpha - 1, \\ w_j &= (j - (N_w - 1)/2)\Delta w \quad j = 0, ..., N_w - 1, \end{aligned}$$

where Δw is the pixel height and $\Delta\alpha D$ is the pixel width. For the projection angles, we consider N_s projections in a uniform rate of N projections per 2π-rotation. s is discretized via:

$$s_k = k\Delta s, \quad \text{where} \quad \Delta s = \frac{2\pi}{N} = \frac{1}{N_s}, \quad k = 0, ..., N_s - 1.$$

Therefore the data will be available as a three-dimensional object with shape (N_s, N_α, N_w).

CF1: Derivative. At constant direction, the derivative term can be expressed as:

$$g_1(s, \alpha, w) = g'(s, \theta_c(s, \alpha, w)) = \lim_{\varepsilon \to 0} \frac{g(s + \varepsilon, \theta_c(s, \alpha, w)) - g(s, \theta_c(s, \alpha, w))}{\varepsilon}$$

There is a big variety of schemes that have been proposed to discretize and compute this view-dependent derivative, each with its advantages and drawbacks [28]. Here we will focus on the two major options.

The standard used for many years was the Noo-Pack-Heurscher (NPH) Scheme. This scheme was suggested shortly after the first publication of

Katsevich by Noo et al. [15] as an improvement of the original direct discretization. It applies the chain rule:

$$g_1(s,\alpha,w) = g'(s,\theta) = \left(\frac{\partial g}{\partial s} + \frac{\partial g}{\partial \alpha}\frac{\partial \alpha}{\partial s} + \frac{\partial g}{\partial w}\frac{\partial w}{\partial s}\right)(s,\alpha,w) = \left(\frac{\partial g}{\partial s} + \frac{\partial g}{\partial \alpha}\right)(s,\alpha,w) \tag{3.2}$$

and then computes the derivatives numerically using central differences. Then for $k = 0, ..., N_s - 2$, $i = 0, ..., N_\alpha - 2$, $j = 0, ..., N_w - 1$:

$$\begin{aligned} g_1(s_{k+1/2},\alpha_{i+1/2},w_j) \simeq\ & \frac{1}{2}\left(\frac{\partial g}{\partial s}(s_{k+1/2},\alpha_{i+1},w_j) + \frac{\partial g}{\partial s}(s_{k+1/2},\alpha_i,w_j)\right) \\ &+ \frac{1}{2}\left(\frac{\partial g}{\partial \alpha}(s_{k+1},\alpha_{i+1/2},w_j) + \frac{\partial g}{\partial \alpha}(s_k,\alpha_{i+1/2},w_j)\right) \\ &= \frac{g_1(s_{k+1},\alpha_i,w_j) - g_1(s_k,\alpha_i,w_j)}{2\Delta s} \\ &+ \frac{g_1(s_{k+1},\alpha_{i+1},w_j) - g_1(s_k,\alpha_{i+1},w_j)}{2\Delta s} \\ &+ \frac{g_1(s_{k+1},\alpha_{i+1},w_j) - g_1(s_k,\alpha_{i+1},w_j)}{2\Delta \alpha} \\ &+ \frac{g_1(s_{k+1},\alpha_{i+1},w_j) - g_1(s_{k+1},\alpha_i,w_j)}{2\Delta \alpha}, \end{aligned} \tag{3.3}$$

where $\alpha_{i+1/2} = \alpha_i + \Delta\alpha/2$ and $s_{k+1/2} = s_k + \Delta s/2$. While this scheme was widely used and accepted, it suffers from the fact that has a non-isotropic resolution; the methods proposed later tried to remedy to this flaw.

More recently, Katsevich proposed the K scheme, a new scheme that is a simpler alternative of the Noo-Hoppe-Dennerlein-Lauritsch-Hornegger (NHDLH) scheme, specifically for curved detector and helical source trajectory [28, 29]. Starting from the derivative of the data:

$$\frac{\partial g}{\partial s}(s,\theta) \simeq \frac{g(s+\delta,\alpha(s+\delta,\theta),w(\theta)) - g(s-\delta,\alpha(s-\delta,\theta),w(\theta))}{2\delta} = g'(s,\alpha,\theta), \tag{3.4}$$

with $\delta > 0$. Let $\hat{\theta}$ be such that $\alpha_{k+1/2} = \alpha(s_i,\hat{\theta})$. The derivative is computed at the points $g'(s_k,\alpha_{i+1/2},w_j)$ using the auxiliary points $(s_i+\delta,\alpha(s_i+\delta,\hat{\theta}))$ and $(s_i-\delta,\alpha(s_i-\delta,\hat{\theta}))$ using a central difference discretization. Let $g_{k,i} = g(s_k,\alpha_i,\cdot)$. The projection data $g(s_i+\delta,\alpha(s_i+\delta,\hat{\theta}))$ is estimated interpolating between the values $g_{k,i}$, $g_{k+1,i}$, $g_{k+1,i+1}$ and $g_{k,i+1}$, and $g(s_i - \delta,\alpha(s_i-\delta,\hat{\theta}))$ is estimated interpolating between $g_{k,i+1}$, $g_{k-1,i+1}$, $g_{k-1,i}$ and

$g_{k,i}$. The interpolation coefficients for these terms are respectively the weights a_1, a_2, a_3, a_4 derived in [29]:

$$a_1 = \frac{1}{2}\left(1 - \frac{\delta}{\Delta\alpha/2}\right) - \frac{\delta}{\Delta s} + a_3, \quad a_2 = \frac{\delta}{\Delta s} - a_3, \quad a_4 = \frac{1}{2}\left(1 + \frac{\delta}{\Delta\alpha/2}\right) - a_3, \tag{3.5}$$

where a_3 is chosen freely (under the condition that all $a_i > 0$). Substituting into Equation (3.4) yields:

$$\begin{aligned} g'(s_k, \alpha_{i+1/2}, w_j) \simeq & \frac{a_1(g_{k,i} - g_{k,i+1}) + a_2(g_{k+1,i} - g_{k-1,i+1})}{2\delta} \\ & + \frac{a_3(g_{k+1,i+1} - g_{k-1,i}) + a_4(g_{k,i+1} - g_{k,i})}{2\delta} \\ = & \frac{1}{2}[r(g_{k+1,i+1} - g_{k-1,i}) + \\ & \left(\frac{1}{\Delta s} - r\right)(g_{k+1,i} - g_{k-1,i+1}) + \\ & \left(\frac{1}{\Delta s} + \frac{1}{\Delta\alpha/2} - 2r\right)(g_{k,i+1} - g_{k,i})], \quad 0 \le r \le \frac{1}{\Delta s}, \end{aligned} \tag{3.6}$$

where $r = \frac{a_3}{\delta}$. For the experiments in this research we set $r = 1$ for simplicity.

CF2: Length-Correction Weighting.

$$g_2(s, \alpha, w) = \frac{D}{\sqrt{D^2 + w^2}} g_1(s, \alpha, w).$$

CF3: Forward Height Rebinning. The field of view (FOV), *i.e.*, the volume of interest inside the helical cylinder, has to be fully illuminated by the fan beam at all rotation angles s. So, the relevant angle is the maximum angle from the source to the edge of the FOV, not to the edge of the detector. Let r be the maximum object radius. The half-fan angle is given by $\alpha_m = \arcsin(r/R)$. Then:

$$g_3(s, \alpha, \psi) = g_2(s, \alpha, w_\kappa(\alpha, \psi)) \quad \text{for all } \psi \in [-\frac{\pi}{2} - \alpha_m, \frac{\pi}{2} + \alpha_m]$$

The interval is the same used in Katsevich [7] using $s_2 = s + 2\psi$ and $\Delta = \pi - 2\alpha_m$:

$$\begin{aligned} & s_2 \in [s - 2\pi + \Delta, s + 2\pi - \Delta] \\ \implies \; & \psi \in [-\pi + \Delta/2, \pi - \Delta/2] \\ \implies \; & \psi \in [-\pi/2 - \alpha_m, \pi/2 + \alpha_m]. \end{aligned}$$

Note that, at fixed ψ, $w_\kappa(\alpha)$ describes a curve in the detector plane $DP(s)$ that is called a $\kappa-$curve of fixed angle ψ. So, $g_3(s, \cdot, \psi)$ represents the projection points on the intersection of the $\kappa-$plane $\kappa(s, \psi)$ with the detector plane $DP(s)$.

After specifying the samples ψ_ℓ, we can implement this with a simple linear interpolation in w. It is recommended to choose:

$$\psi_\ell = \ell \Delta\psi = \ell \frac{\pi + 2\alpha_m}{2M}, \quad \ell = -M, ..., M.$$

Noo [15] suggests to choose M such that:

$$\begin{aligned} \frac{\mathrm{d}w_\kappa}{\mathrm{d}\psi}(-\alpha_m, \ \pi/2 + \alpha_m) &= \frac{Dh}{R}\left(\cos(\alpha_m) + \sin(\alpha_m)\left(\tan(\alpha_m) + \frac{\alpha_m + \pi/2}{\cos^2(\alpha_m)}\right)\right) \\ &\simeq \frac{d_w}{\Delta\psi} = \frac{2Md_w}{\pi + 2\alpha_m}. \end{aligned}$$

Note that to avoid unnecessary calculations, here we used the definition of w_κ for constant pitch. Then we get that the number of filtering lines M is:

$$M = \left\lfloor \left(\frac{\pi}{2} + \alpha_m\right) \frac{DP}{2Rd_w} \left(\cos(\alpha_m) + \sin(\alpha_m)\left(\tan(\alpha_m) + \left(\alpha_m + \frac{\pi}{2}\right)\frac{\sin(\alpha_m)}{\cos^2(\alpha_m)}\right)\right)\right\rfloor. \tag{3.7}$$

CF4: 1D Hilbert Transform. Compute, for a fixed ψ,

$$\begin{aligned} g_4(s, \alpha, \psi) &= \int_{-\pi/2}^{\pi/2} k_H(\sin(\alpha - \alpha')) g_3(s, \alpha', \psi)\, d\alpha' \\ &= (k_H(\sin(\cdot)) * g_3(s, \cdot, \psi))(\alpha). \end{aligned}$$

This step is very similar to standard fan-beam CT. The kernel h_H can be approximated in the following way. Let $H(t) = \frac{t\Delta\alpha}{\sin(t\Delta\alpha)} \int_0^1 \sin(\pi t u) A(u)\, du$. Then:

$$k_H(\sin\alpha) = \frac{\alpha}{\sin\alpha} k_H(\alpha) \simeq \frac{1}{\Delta\alpha} H\left(\frac{\alpha}{\Delta\alpha}\right) = \frac{1}{\Delta\alpha}\frac{\alpha}{\sin(\alpha)} \int_0^1 \sin\left(\frac{\pi\alpha u}{\Delta\alpha}\right) A(u)\, du,$$

where $A(u)$ is the apodization window, used to reduce streak artifacts and oscillations. Thus we obtain the following discrete convolution formula:

$$g_4(s,\alpha_i,\psi_\ell) \simeq \sum_{p=0}^{N_\alpha-1} H(i-(p+1/2))g_3(s,\alpha_{p+1/2},\psi_\ell).$$

Note that the shift that was introduced in step 1 is brought back to the original position.

CF5: Backward Height Rebinning. Compute:

$$g_5(s,\alpha,w) = g_4(s,\alpha,\hat{\psi}(\alpha,w)), \tag{3.8}$$

where $\hat{\psi}$ is the angle with the smallest absolute value that satisfies:

$$w_\kappa(\alpha) = \frac{Dh}{R}\left(\psi\cos\alpha + \frac{\psi}{\tan\psi}\sin\alpha\right), \qquad \text{(constant pitch)}$$

$$w_\kappa(\alpha,s) = -\frac{D\cos(\alpha)\mathbf{e}_v(s)\cdot n(s,\psi) + D\sin(\alpha)\mathbf{e}_u(s)\cdot n(s,\psi)}{\mathbf{e}_w\cdot n(s,\psi)} \qquad \text{(variable pitch).}$$

We can get this using simple linear interpolation:

$$g_5(s,\alpha_i,w) \simeq (1-c(\alpha_i,w,\ell))g_4(s,\alpha_i,\psi_\ell) + c(\alpha_i,w,\ell)g_4(s,\alpha_i,\psi_{\ell+1}), \tag{3.9}$$

$$c(\alpha_i,w,\ell) = \frac{w - w_\kappa(\alpha_i,\psi_\ell)}{w_\kappa(\alpha_i,\psi_{\ell+1}) - w_\kappa(\alpha_i,\psi_\ell)}, \tag{3.10}$$

for w such that

$$w_\kappa(\alpha_i,\psi_\ell) < w < w_\kappa(\alpha_i,\psi_{\ell+1}).$$

For constant pitch, we use the following logic:

- For $\alpha_i \geq 0$, compute $g_5(s,\alpha_i,w_j)$ from $\ell = -M$, and increase ℓ until w_j satisfies:

  $$w_\kappa(\alpha_i,\psi_\ell) < w_j < w_\kappa(\alpha_i,\psi_{\ell+1});$$

  So basically we are moving the window $(w_\kappa(\psi_\ell), w_\kappa(\psi_{\ell+1}))$ until w_j is within this window.

- For $\alpha_i < 0$, compute $g_5(s,\alpha_i,w_j)$ from $\ell = M$, and decrease ℓ until w_j

satisfies:

$$w_\kappa(\alpha_i, \psi_{\ell-1}) < w_j < w_\kappa(\alpha_i, \psi_\ell);$$

This logic assumes that the mapping from $\kappa-$line element ψ to detector row position w_κ is monotonic in ψ. So it searches for the first interval $[w_\kappa(\psi_\ell), w_\kappa(\psi_{\ell+1})]$, that contains a given detector row value w, and stops there. This works only if w increases or decreases smoothly with ψ.

But when w_κ is not monotonic in ψ - for example, it bends, loops, or flattens - the same detector row w can lie inside multiple intervals. This logic then picks the wrong one because it just stops at the first match.

For variable pitch, we have no guarantee that the $\kappa-$lines are monotonic. Therefore we implement a new logic that checks all intervals and picks the one where w is closest to the middle of the interval. This ensures it always finds the most relevant $\kappa-$line, even if the $\kappa-$lines are not monotonic.

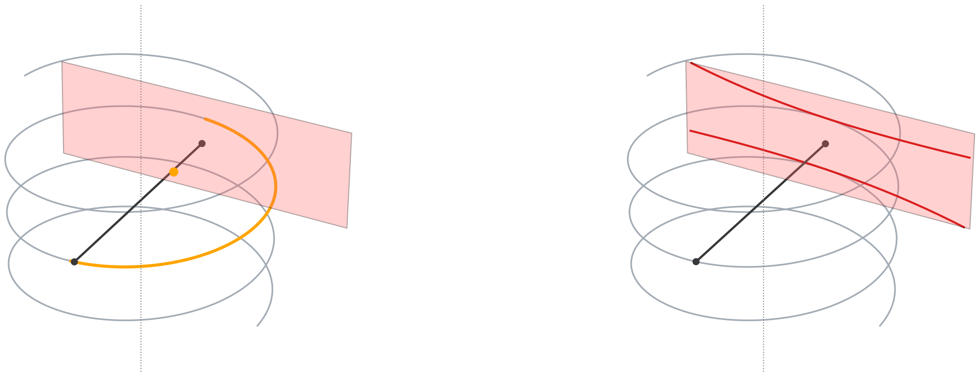
CF6: Post Cosine weighting.

$$g^F(s, \alpha, w) = \cos\alpha g_5(s, \alpha, w) \tag{3.11}$$

TD Windowing. One of the main challenges in implementing the Katsevich algorithm is that the integration interval for the backprojection step depends on the reconstruction point x, since it must be limited to the corresponding $PI-$interval $I_{\text{PI}}(x)$. Explicitly computing this interval for each point is computationally expensive and makes the algorithm less modular.

To overcome this, we exploit a key geometric result stated by Noo et al. in [30]: integrating over $PI-$lines can be substituted with masking the data using the Tam–Danielsson window [25], and this still provides the minimum amount of data for exact reconstruction. This implies that we can perform backprojection over the entire helical trajectory, provided that for each source position s, we restrict the detector data to the TD window. This way we effectively limit the integration to the relevant $PI-$interval for each x, without explicitly computing it. Figure 3.1 illustrates the difference between the two strategies.

The TD window $TD(s)$ consists of the region bounded by the projection of one full helical turn above and below the current source point, and so it depends on the geometry of the helix and detector, see the illustration in Figure 3.2.



(a) Bounded integration: only a limited segment of the helix (yellow) contributes, determined by the reconstruction point x (yellow dot).

(b) Unbounded integration with TD windowing: the TD window is applied over the entire helix, independent of the reconstruction point.

Figure 3.1: Comparison between bounded and unbounded integration in helical CT. In (a), the integration range depends on the target point x, requiring recalculation for each location. In (b), considering only the data within the TD window enables integration over the full helix, making the operation independent of the reconstruction point and applicable globally.

Let $\chi(\alpha, w)$ denote the characteristic function of $TD(s)$, defined over detector coordinates (α, w); this function is used to mask out irrelevant detector data before backprojection.

For constant pitch, this region is defined as the set of points $\{\alpha, w\}$ such that:

$$\alpha \in [-\alpha_m, \alpha_m],$$
$$w_{\text{bottom}}(\alpha) \leq w \leq w_{\text{top}}(\alpha),$$

where the projections of lower and upper turn of the helix on the detector plane $DP(s)$ are given respectively by:

$$w_{\text{bottom}}(\alpha) = -\frac{DP}{2\pi R}\frac{\alpha + \pi/2}{\cos\alpha}, \tag{3.12}$$
$$w_{\text{top}}(\alpha) = \frac{DP}{2\pi R}\frac{-\alpha + \pi/2}{\cos\alpha} \tag{3.13}$$

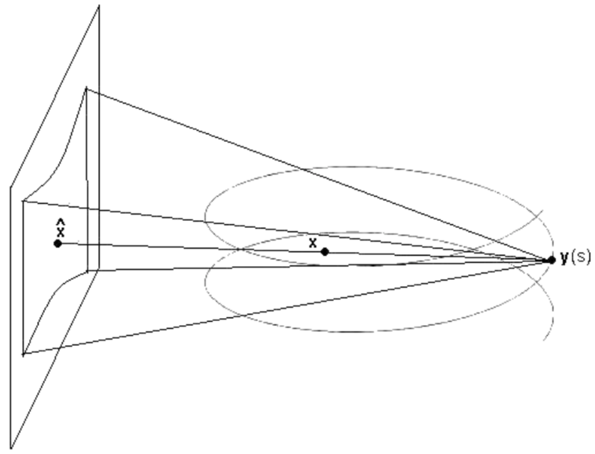


Figure 3.2: Tam–Danielsson window $TD(s)$.

For variable pitch $p(s)$, the matter becomes more difficult as these curves are not equal for all projections but depend on the local configuration of the helix at each s_0. Ye *et al.* [14] derives the expressions for α and w parametrized by s. The generalized versions for a variable pitch are:

$$\alpha(s) = \frac{D \sin(s - s_0)}{1 - \cos(s - s_0)}, \tag{3.14}$$

$$w(s) = \frac{D}{R} \frac{p(s) - p(s_0)}{1 - \cos(s - s_0)}, \tag{3.15}$$

with $s \in [s_0 - 2\pi + \arccos(r/R), s_0 - \arccos(r/R)]$ for the lower projection and $[s_0 + \arccos(r/R), s_0 + 2\pi - \arccos(r/R)]$ for the upper projection, where r is the radius of the object's field of view and R is the source-to-origin distance. The derivation of these equations can be found in Appendix C.

In practice, we need the smooth counterpart of χ. Noo *et al.* [15] suggest to use:

$$\chi_a(s, \alpha, w) = \begin{cases} 0, & \text{if } w > w_{\text{top}}(\alpha) + ad_w, \\ \dfrac{w_{\text{top}}(\alpha, s) + ad_w - w}{2ad_w}, & \text{if } w_{\text{top}}(\alpha, s) - ad_w < w \leq w_{\text{top}}(\alpha, s) + ad_w, \\ 1, & \text{if } w_{\text{bottom}}(\alpha, s) + ad_w < w \leq w_{\text{top}}(\alpha, s) - ad_w, \\ \dfrac{w - w_{\text{bottom}}(\alpha, s) + ad_w}{2ad_w}, & \text{if } w_{\text{bottom}}(\alpha, s) - ad_w < w \leq w_{\text{bottom}}(\alpha, s) + ad_w, \\ 0, & \text{if } w \leq w_{\text{bottom}}(\alpha, s) - ad_w. \end{cases}$$

where a is a smoothing factor. Note that for the constant pitch case,

$$w_{\text{bottom}}(\alpha, s) = w_{\text{bottom}}(\alpha),$$
$$w_{\text{top}}(\alpha, s) = w_{\text{top}}(\alpha).$$

Multiplying the filtered projections with this characteristic function, we obtain:

$$g^{BP}(s_k, \alpha_i, w_j) = \chi_a(s_k, \alpha_i, w_j) g^F(s_k, \alpha_i, w_j). \tag{3.16}$$

3.1.2 Katsevich Algorithm for Flat Detector

While the formulas may differ, the implementation strategies are the same for the two geometries. Therefore, for flat detector we only briefly mention the steps of the algorithm; for more details regarding the numerical strategies, refer to the curved detector case in Section 3.1.1.

The detector coordinates are discretized via:

$$u_i = (i - (N_u - 1)/2)\Delta u \quad i = 0, \ldots, N_u - 1,$$
$$w_j = (j - (N_w - 1)/2)\Delta w \quad j = 0, \ldots, N_w - 1,$$

where Δw is the pixel height and Δu is the pixel width. For the projection angles, we consider N_s projections in a uniform rate of N projections per 2π-rotation. s is discretized via:

$$s_k = k\Delta s, \quad \text{where} \quad \Delta s = \frac{2\pi}{N} = \frac{1}{N_s}, \quad k = 0, \ldots, N_s - 1.$$

As for the curved case, the data will be available as a three-dimensional object with shape (N_s, N_u, N_w).

FF1: Derivative. The derivative in Theorem 3 can be computed via chain rule. The detector has $N_w \times N_u$ pixels. Discretize the detector coordinates via:

$$u_i = (i - 1 - \frac{N_u}{2})\Delta u \quad i = 1, \ldots, N_u,$$
$$w_j = (j - 1 - \frac{N_w}{2})\Delta w \quad w = 1, \ldots, N_w.$$

Moreover s is discretized as s_k for some range of $k = 1, ..., K$. Noo et al. [15] suggests to implement the derivative using the chain rule as follows:

$$
\begin{aligned}
g_1(s_{k+1/2}, u_{i+1/2}, w_{j+1/2}) \simeq & \sum_{m=i}^{i+1} \sum_{n=j}^{j+1} \frac{g(s_{k+1}, u_m, w_n) - g(s_k, u_m, w_n)}{4\Delta s} \\
& + \left(\frac{u_{i+1/2}^2 + D^2}{D} \right) \sum_{p=k}^{k+1} \sum_{n=j}^{j+1} \frac{g(s_p, u_{i+1}, w_n) - g(s_p, u_i, w_n)}{4\Delta u} \\
& + \left(\frac{u_{i+1/2} w_{j+1/2}}{D} \right) \sum_{p=k}^{k+1} \sum_{m=i}^{i+1} \frac{g(s_p, u_m, w_{j+1}) - g(s_p, u_m, w_j)}{4\Delta w}.
\end{aligned}
$$

Alternatively, the K scheme was adapted by Katsevich to work also for flat detector. The additional difficulty using this method for flat detector is that also w depends on s. The problem was tackled by using a partially optimized chain rule scheme:

$$
\begin{aligned}
g_1(s, u, w) &= \frac{d}{ds} g(s, u(s, \theta), w(s, \theta)) \\
&= \frac{d}{ds} g(s, u(s, \theta), W)\Big|_{W = w(s,\theta)} + \frac{\partial g}{\partial w}(s, u(s, \theta), w(s, \theta)) \frac{\partial w}{\partial s}(s, \theta).
\end{aligned} \tag{3.17}
$$

The chain rule is applied only with respect to w. In the first term w is assumed to be constant, so if we compute $g(s_k, u_{i+1/2}, w_j)$ we do not need any interpolation along w.

Now follow the same approach as in 3.4, 3.5, 3.6. Assuming that $u(s, \hat{\theta})$ can be approximated to a straight line with slope $u'_s(s_k, \hat{\theta})$, the following formula is obtained:

$$
\begin{aligned}
g_1(s_k, u_{i+1/2}, w_j) \simeq & \frac{1}{2}[r(g_{k+1,i+1,j} - g_{k-1,i,j}) + \\
& \left(\frac{1}{\Delta s} - r \right) (g_{k+1,i,j} - g_{k-1,i+1,j}) + \\
& \left(\frac{1}{\Delta s} + \frac{u'_s}{\Delta u/2} - 2r \right) (g_{k,i+1,j} - g_{k,i,j})], \quad 0 \le r \le \frac{1}{\Delta s},
\end{aligned} \tag{3.18}
$$

where again, for simplicity $r = 1$ is used.

FF2: Length Correction Weighting.

$$g_2(s,u,w) = \frac{D}{\sqrt{u^2 + D^2 + w^2}} g_1(s,u,w). \tag{3.19}$$

FF3: Forward Height Rebinning. As seen in the curved detector case, the half-fan angle is given by $\alpha_m = \arcsin(r/R)$. X-rays beyond this angle won't intersect the object's field of view. Using linear interpolation, we compute:

$$g_3(s,u,w) = g_2(s,u,w_\kappa(u,\psi)) \quad \forall \psi_n \in [-\frac{\pi}{2} - \alpha_m, \frac{\pi}{2} + \alpha_m]. \tag{3.20}$$

FF4: 1D Hilbert Transform. At constant ψ, compute

$$g_4(s,u,\psi) = \int_{-\infty}^{\infty} k_H(u - u') g_3(s,u',\psi) du'.$$

Since we have a discrete set of data we need the discrete formulation of the transform. From the Hilbert transform kernel formula with cut-off frequency b:

$$\begin{aligned} k_H(u) &\simeq -\int_{-b}^{b} i\text{sgn}(\xi) e^{12\pi\xi u}\, d\xi \\ &= \frac{1}{\pi u}[1 - \cos(2\pi b u)]. \end{aligned}$$

Sample u by $u_n = n\Delta u$. To avoid aliasing errors, we satisfy the Nyquist condition by setting $b = \frac{1}{2\Delta u}$ [31]. Then we obtain the discrete kernel:

$$k_H[n] = \frac{1}{\pi n \Delta u}[1 - \cos(\pi n)]. \tag{3.21}$$

The discrete Hilbert transform therefore becomes:

$$g_4(s,u_n,\psi_\ell) = \Delta u \sum_k k_H[n-k] g_3(s,u_k,\psi_\ell) = \sum_k \frac{1}{\pi n}[1 - \cos(\pi n)] g_3(s,u_k,\psi_\ell), \tag{3.22}$$

which is a discrete convolution that can be implemented with FFT.

FF5: Backward Height Rebinning. In this step the data points are mapped from the $\kappa-$lines back to the detector grid.

$$g^F(s,u,w) = g_4(s,u,\hat{\psi}(u,w)),$$

where $\hat{\psi}(u,w)$ is the angle ψ with the smallest absolute value such that it satisfies:

$$w_\kappa(u) = \frac{Dh}{R}\left(\psi + \frac{\psi}{D\tan\psi}u\right) \quad \text{(constant pitch)},$$
$$w_\kappa(u) = -\frac{D\mathbf{e}_v(s)\cdot n(s,\psi) + u\mathbf{e}_u(s)\cdot n(s,\psi)}{\mathbf{e}_w\cdot n(s,\psi)} \quad \text{(variable pitch)}.$$

The strategy used is the same as for the curved case.

TD Windowing. The Tam-Danielsson window is applied to the projections before backprojecting. Let $\chi(u,w)$ be the characteristic function of the Tam-Danielsson window.

For constant pitch, this region is defined as the set of points $\{u,w\}$ such that:

$$u \in [-D\tan\alpha_m, D\tan\alpha_m],$$
$$w_{\text{bottom}}(u) \leq w \leq w_{\text{top}}(u).,$$

where the projections of lower and upper turn of the helix on the detector plane $DP(s)$ are given respectively by:

$$w_{\text{bottom}}(u) = -\frac{P}{2\pi RD}(u^2 + D^2)\arctan(u/D) + \pi/2, \tag{3.23}$$
$$w_{\text{top}}(u) = \frac{P}{2\pi RD}(u^2 + D^2)\arctan(u/D) - \pi/2. \tag{3.24}$$

For variable pitch $p(s)$, we have u and w parametrized by s:

$$u(s) = D\tan\left(\frac{\sin(s-s_0)}{1-\cos(s-s_0)}\right), \tag{3.25}$$
$$w(s) = \frac{D}{R}\frac{p(s)-p(s_0)}{1-\cos(s-s_0)}, \tag{3.26}$$

with $s \in [s_0 - 2\pi + \arccos(r/R), s_0 - \arccos(r/R)]$ for the lower projection and $[s_0 + \arccos(r/R), s_0 + 2\pi - \arccos(r/R)]$ for the upper projection, where r is the radius of the object's field of view.

Similarly to the curved case we actually consider its smooth counterpart:

$$\chi_a(s,u,w) = \begin{cases} 0, & \text{if } w > w_{\text{top}}(u) + ad_w, \\ \frac{w_{\text{top}}(u,s)+ad_w-w}{2ad_w}, & \text{if } w_{\text{top}}(u,s) - ad_w < w \le w_{\text{top}}(u,s) + ad_w, \\ 1, & \text{if } w_{\text{bottom}}(u,s) + ad_w < w \le w_{\text{top}}(u,s) - ad_w, \\ \frac{w-w_{\text{bottom}}(u,s)+ad_w}{2ad_w}, & \text{if } w_{\text{bottom}}(u,s) - ad_w < w \le w_{\text{bottom}}(u,s) + ad_w, \\ 0, & \text{if } w \le w_{\text{bottom}}(u,s) - ad_w \end{cases} \tag{3.27}$$

where a is a smoothing factor. Note that for the constant pitch case, $w_{\text{bottom}}(u,s) = w_{\text{bottom}}(u)$ and $w_{\text{top}}(u,s) = w_{\text{top}}(u)$.
Multiplying the filtered projections with this characteristic function we obtain:

$$g^{BP}(s_k, u_i, w_j) = \chi_a(s_k, u_i, w_j) g^F(s_k, u_i, w_j). \tag{3.28}$$

3.2 Implementation in ODL

The focus of this thesis is on the theoretical foundations and methodology of tomographic reconstruction using Katsevich's formula, but a substantial part of the work has been devoted to its practical implementation in the ODL framework. To maintain the narrative flow, detailed code explanations are deferred to Appendix D, which provides an in-depth account of the design of the pre-existing ODL operators used in tomographic reconstruction, including the Radon transform, the FBP pipeline, and the acquisition geometry setup, together with the algorithmic details of the newly implemented operators for Katsevich's method. Here we summarize only the most relevant aspects of the Katsevich reconstruction algorithm design.

Following ODL's modular design, the Katsevich reconstruction has been implemented as the composition of three custom operators: the filtering operator $\mathcal{D}(\cdot)$, the Tam–Danielsson (TD) window operator $TD(\cdot)$, and the weighted adjoint operator $\mathcal{R}_w^*(\cdot)$.

Filtering operator. Implemented in two variants, `KatsevichFilterCurved` and `KatsevichFilterFlat`, this operator handles both curved and flat detector geometries. Upon initialization, all geometry-dependent quantities are extracted from the input `ray_trafo` and the forward and backward rebinning maps are precomputed, since they depend only on the acquisition geometry and not on the data. This allows them to be reused for multiple reconstructions without recomputation.

TD window operator. This operator applies a mask to the projection data, removing contributions outside the Tam–Danielsson window. In the constant pitch case the bounds of the window are fixed and computed once from the acquisition parameters, whereas in the variable pitch case they are recalculated for each view to remain consistent with the changing geometry.

Weighted adjoint operator. The last operator, $\mathcal{R}_w^*$, is implemented as a modification of ODL's standard adjoint operator. The Katsevich backprojection requires applying, inside the backprojection integral, a voxel- and source position-dependent weight:

$$\frac{1}{v^*(s, \mathbf{x})} = \frac{1}{R - \langle \mathbf{x}, \mathbf{e}_v(s) \rangle}, \tag{3.29}$$

where R is the source–origin distance and $\mathbf{e}_v(s)$ the unit vector along the source-detector center direction. Because this term couples the voxel position $\mathbf{x}$ and the projection angle s, it cannot be applied as a simple post-processing step after a global backprojection. The standard ODL backprojection method relies on the ASTRA backend which does not currently allow passing such a custom weight function.

To address this, the modified operator performs the backprojection source view by source view. For each projection angle s_k, a backprojection is computed with the ASTRA backend, and the weighting term (3.29) is evaluated for that s_k and applied element-wise to the resulting volume before accumulation.

As a final note, while the general implementation supports both flat and curved detector geometries, the focus of this work was on curved detectors, the standard detector used in clinical CT systems. In particular, support for general variable pitch trajectories is implemented only in the curved case. The flat detector implementation is kept as a simplified case used primarily for validation and benchmarking purposes

Chapter 4

Experimental Setup

4.1 Image Quality Metrics for Reconstruction Evaluation

In some of the experiments conducted, the original object (phantom) is synthetically generated and thus fully known. This enables a direct and quantitative comparison between the reconstructed image and the ground truth. To assess the fidelity, we adopt two widely used image quality metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). These metrics are complementary: PSNR provides a quantitative, voxel-wise measure of error, while SSIM evaluates the perceptual and structural similarity between images. PSNR expresses reconstruction quality on a logarithmic decibel scale:

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\max_O^2}{\mathrm{MSE}(O,R)}\right), \tag{4.1}$$

where $\max_O$ is the maximum voxel intensity in the reference image O, and MSE is the mean squared error between O and the reconstruction R. Higher PSNR values indicate better reconstructions . PSNR is popular for quantifying the pixel-wise intensity error, however it does not consider the spatial or perceptual structure of the images. To address this, we introduce the Structural Similarity Index Measure (SSIM), which compares luminance, contrast, and structural information:

$$\mathrm{SSIM}(O,R) = \frac{(2\mu_O\mu_R + C_1)(2\sigma_{OR} + C_2)}{(\mu_O^2 + \mu_R^2 + C_1)(\sigma_O^2 + \sigma_R^2 + C_2)}, \tag{4.2}$$

where μ, σ, and σ_{OR} are respectively local means, variances, and covariance. SSIM ranges from 0 (no similarity) to 1 (perfect match) and is widely used in CT to assess perceptual fidelity. The combination of PSNR and SSIM provides a robust framework to assess both quantitative accuracy and structural quality in the reconstructions performed.

4.2 Datasets

To assess the performance and robustness of the reconstruction methods, we consider three datasets of increasing complexity. The progression from synthetic to clinical data reflects a gradual shift from fully controlled, idealized conditions to realistic acquisition scenarios.

The first dataset consists of fully synthetic projections generated from a known phantom. This setting offers complete control over the geometry and the absence of noise, making it possible to evaluate reconstruction accuracy against a ground truth.

The second dataset contains experimental measurements from wooden logs. This setup introduces more realistic acquisition conditions such as measurement noise, high-resolution detectors, and variable pitch.

Finally, the third dataset consists of real clinical CT scans, the main interest of our study. While the wooden log data already included measurement noise and non-ideal geometry, the clinical scans add further challenges due to the wide range of tissue types present in the human body and the simulation of advanced acquisition techniques such as flying focal spot (FFS), which improves resolution but increases reconstruction complexity.

In the following sections, we introduce these datasets and provide details of the acquisition geometry, object characteristics, and any dataset-specific adaptations to the reconstruction pipeline.

4.2.1 Synthetic Data

The synthetic data are generated by creating a known phantom and simulating its projection data using a ray transform. The phantom chosen for the testing is the well-known Shepp–Logan phantom. Its complex structure and sharp intensity transitions present a suitable challenge for the reconstruction methods, while the synthetic setting ensures perfectly noise-free images, allowing for an unbiased assessment of reconstruction accuracy.

Volume Shape ($N_{\text{cols}} \times N_{\text{rows}} \times N_{\text{slices}}$)	$512 \times 256 \times 300$
Voxel size	0.1 mm
Detector pixel size	0.15625 mm
Detector Rows (N_w)	250
Detector Columns (N_u or N_α)	275
Detector Curvature Radius (ρ)	130 mm
Source Positions per Turn	1056
Source to Origin Distance (R_s)	50 mm
Source to Detector Distance (D)	80 mm
Helical Pitch (P)	40.43 mm/turn

Table 4.1: Acquisition geometry parameters for the synthetic dataset.

For these experiments, a fixed acquisition geometry is used, with parameters summarized in Table 4.1. The projection data are generated for both flat and curved detector configurations. The geometry settings are kept constant across all experiments with synthetic data.

4.2.2 Wooden Logs Data

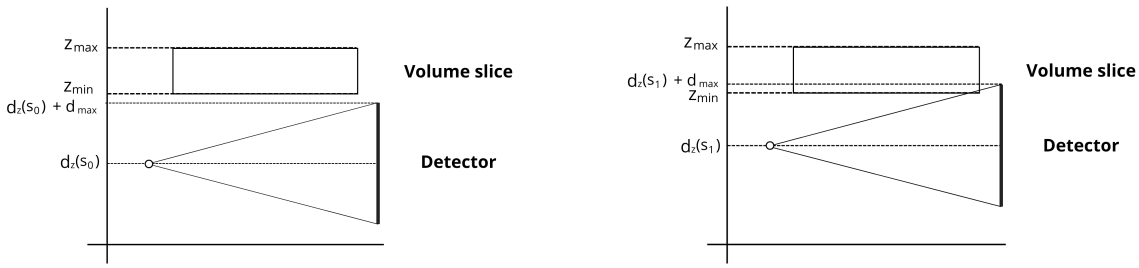
Moving from the idealized synthetic setup to more realistic scenarios, we consider experimental data that are subject to measurement noise and are acquired under non-ideal geometric conditions, such as a variable pitch. These factors introduce additional complexity compared to the synthetic case. Note that since no known ground truth is available for direct comparison, the evaluation will focus on qualitative inspection and comparison with a standard FBP reconstruction.

This dataset originates from a parallel project aimed at imaging the internal structure of wooden logs in order to detect hidden nodules, which are early formations of branches that weaken the structural integrity of the wood. The CT-based approach offers a non-destructive method to visualize internal defects that are not visible from the outside.

Volume Shape ($N_{\text{cols}} \times N_{\text{rows}} \times N_{\text{slices}}$)	$512 \times 512 \times 738$
Voxel Size	0.5 mm
Volume X Range	$[-80,\ 130]$ mm
Volume Y Range	$[-200,\ 20]$ mm
Volume Z Range	$[0,\ 369.36]$ mm
Detector Columns (N_α)	1560
Detector Rows (N_w)	230
Detector X Range	$[-0.465,\ 0.465]$ mm
Detector Z Range	$[-34.5,\ 34.5]$ mm
Detector Curvature Radius (ρ)	503 mm
Source to Origin Distance (R_s)	503 mm
Source to Detector Distance (D)	503 mm
Helical Pitch (P)	101 mm/turn

Table 4.2: Acquisition geometry parameters for the wooden log dataset.

We are provided with the acquisition parameters and the measured sinogram of the object; the parameters are summarized in Table 4.2. In this setup, the detector has extremely high resolution along the horizontal axis, which poses computational challenges in terms of memory and processing time. To address this, the reconstruction is performed one slice at a time, selecting only the projection views where at least one ray intersects the target slice (as illustrated in Figure 4.1).



(a) Projection view not considered.

(b) Projection view considered.

Figure 4.1: Selection of projection views contributing to the reconstruction of a given slice.

4.2.3 Clinical Data

Finally, we moved on to testing Katsevich algorithm's performance on real clinical scans. Testing the Katsevich formula on clinical data is the ultimate goal of our study, while previous datasets served mainly for validation. Here, several complications are introduced: acquisition is often noisy, pitch may vary across slices, and techniques such as flying focal spot, used to enhance resolution, add additional complexity to the reconstruction problem.

Access to clinical CT data is limited due to patient privacy regulations and vendor-specific formats, which often make the data difficult to share or interpret. A notable exception is the dataset released by the Mayo Clinic, which has become a benchmark in low-dose CT research. The Mayo dataset provides anonymized clinical scans along with both full-dose and simulated low-dose projection data, all made available in an open, vendor-neutral format.

Mayo dataset. This dataset was originally prepared for the *2016 Low Dose CT Grand Challenge*, sponsored by the AAPM, NIBIB, and Mayo Clinic. It includes 30 contrast-enhanced abdominal CT scans, with projection data released for both the full and simulated low-dose acquisitions [32]. Projection data is encoded in the DICOM-CT-PD format, an extended DICOM structure designed specifically for CT projection storage [33]. This format includes the raw projection data and all acquisition geometry needed for reconstruction.

The full dataset, as released on The Cancer Imaging Archive (TCIA), includes 99 head scans (labeled N for neuro), 100 chest (labeled C for chest), and 100 abdomen scans (labeled L for liver), making it one of the most comprehensive open datasets available for CT research. Each scan is accompanied by DICOM-CT-PD projection data, reconstructed DICOM image data, and clinical metadata reports in Excel format. All reconstructions were originally performed using a standard filtered backprojection pipeline.

The scans were acquired on two types of scanners: 50 cases per region (head, chest, abdomen) from a Siemens SOMATOM Definition Flash, and the remaining cases from a GE Lightspeed VCT scanner. The low-dose versions were generated using dose-reduction simulation: abdomen and head scans are provided at 25% of routine dose, and chest scans at 10%. This enables researchers to evaluate algorithm robustness at various dose levels.

Flying focal spot. The projection data in the Mayo dataset were acquired using a single focal spot. However, in our experiments we also investigate the potential impact of the flying focal spot (FFS) technique by slightly modifying the acquisition geometry. In modern CT scanners, FFS alternates the focal spot between two nearby positions $y(s)$ and $y'(s)$ at each view, where:

$$y(s) = DS(s) - D\mathbf{e}_v(s),$$
$$y'(s) = DS(s) - (D + \delta w)\mathbf{e}_v(s) - \delta_u \mathbf{e}_u(s) + \delta_w \mathbf{e}_w,$$

with $\delta_u, \delta_v, \delta_w$ small constants. This technique increases the sampling density along the longitudinal (z) axis without adding detector rows or extending scan time and can improve resolution in z and reduce helical artifacts; however, it also requires modeling two source positions per view in the acquisition geometry.

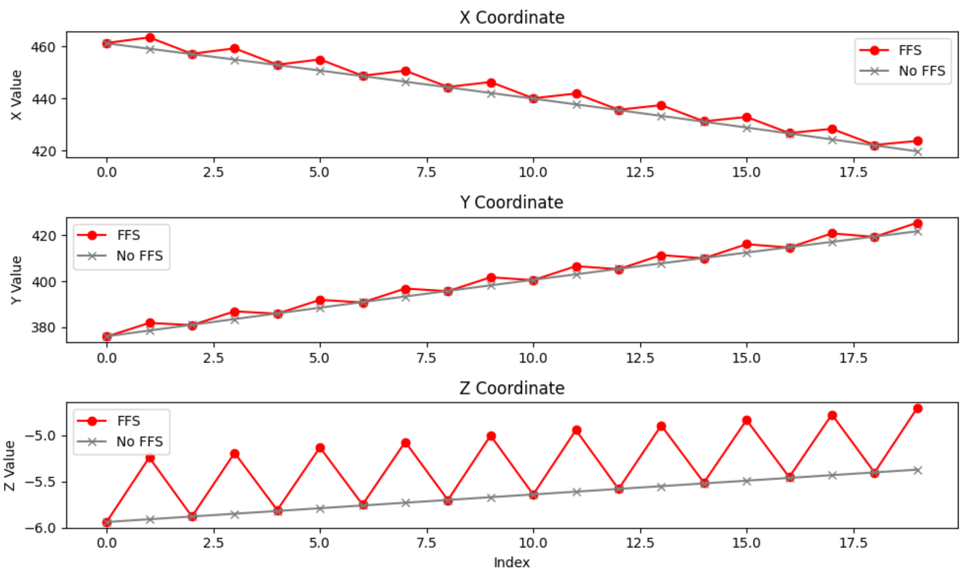


Figure 4.2: $x, y, z-$coordinates of the positions of the source on the helical trajectory with and without using FFS. The positions of the source using FFS are simply obtained by imposing a shift to every second point.

In the ODL implementation, FFS simulation is enabled in the data loader by shifting every second source position by fixed offsets in the $\mathbf{e}_u$, $\mathbf{e}_v$, and $\mathbf{e}_w$ directions. This produces two interlaced helical source trajectories. Instead of doubling the number of projections, we treat each pair as a single view with two focal spot positions. So, for example, 8000 recorded projections become 4000 views, each associated with two source locations. This procedure does not change the sinogram values themselves, since ODL stores the explicit source positions in its geometry object. Nevertheless, having two focal spot positions per view effectively simulates a scan that is twice as fast, a factor that should be considered when evaluating or designing reconstruction algorithms.

The only step in our reconstruction pipeline that currently requires special handling when FFS is used is the Tam–Danielsson (TD) windowing. Ideally, separate TD windows should be computed for the two helices, but currently we apply the mask from the first helix to all positions. While this approximation works reasonably well, performing an exact windowing would likely improve reconstruction quality.

The acquisition geometry is common for all the scans in the dataset, but the volume size, the pitch and number of projections varies for each scan type (abdomen, chest, and head); the parameters relative to the acquisition geometry setups are shown in Table 4.3.

Volume Shape ($N_{\text{cols}} \times N_{\text{rows}} \times N_{\text{slices}}$)	chest: $512 \times 512 \times 351$ abdomen: $512 \times 512 \times 169$
Volume X Range	$[-200,\ 200]$ mm
Volume Y Range	$[-200,\ 200]$ mm
Volume Z Range	chest: $[33.35,\ 384.35]$ mm abdomen: $[272.25,\ 779.25]$ mm
Detector Columns (N_α)	736
Detector Rows (N_w)	64
Detector X Range	$[-0.4359,\ 0.4359]$ mm
Detector Z Range	$[-35.0311,\ 35.0311]$ mm
Detector Curvature Radius (ρ)	1085.6 mm
Source to Origin Distance (R_s)	595 mm
Source to Detector Distance (D)	1085.6 mm
Helical Pitch (P)	chest: 34.44 mm/turn abdomen: 22.98 mm/turn

Table 4.3: Parameters of the acquisition geometry for the scans in the Mayo Clinic dataset. The only parameters changing between chest and abdomen scans are the helical pitch and the volume size in the z-direction.

Chapter 5

Results and Analysis

This chapter presents and discusses the results of the reconstruction using the Katsevich method. We compare the performance to the standard FBP-type method already implemented in ODL under a range of acquisition settings. The code implemented and employed for these experiments can be found in the GitHub repository odl-katsevich under the branch ”curved”.

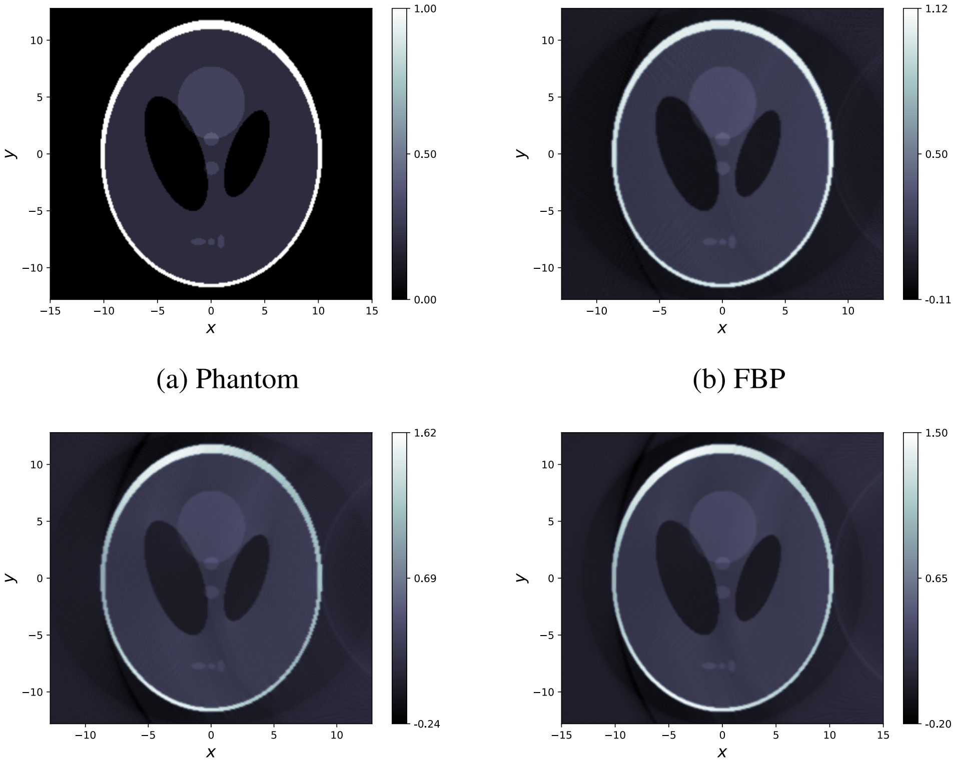
Section 5.1 reports results obtained on synthetic data generated under idealized conditions: noiseless projections and constant pitch. Section 5.2 introduces reconstructions from real CT scans data, including noise and variable pitch. Finally, Section 5.3 presents results on clinical datasets acquired with complex geometries involving advanced techniques such as flying focal spot.

5.1 Synthetic Data

In the first part of our experimental analysis, three reconstruction algorithms are applied to the synthetic dataset: FBP and Katsevich using the two differentiation schemes described in Section 3.1.1. The quality of the reconstructions is assessed using PSNR and SSIM metrics, as defined in Section 4.1, and the relative performance of the methods is analyzed under different levels of angular sparsity.

A visual comparison is shown in Figure 5.1 and Figure 5.2, while quantitative results are reported in Table 5.1 and illustrated in Figure 5.4.

Consider first the flat detector configuration. Figure 5.1 shows the middle slice of the Shepp-Logan phantom and its reconstructions using standard FBP, Katsevich K, and Katsevich NPH. Although the phantom is normalized to the



(a) Phantom

(b) FBP

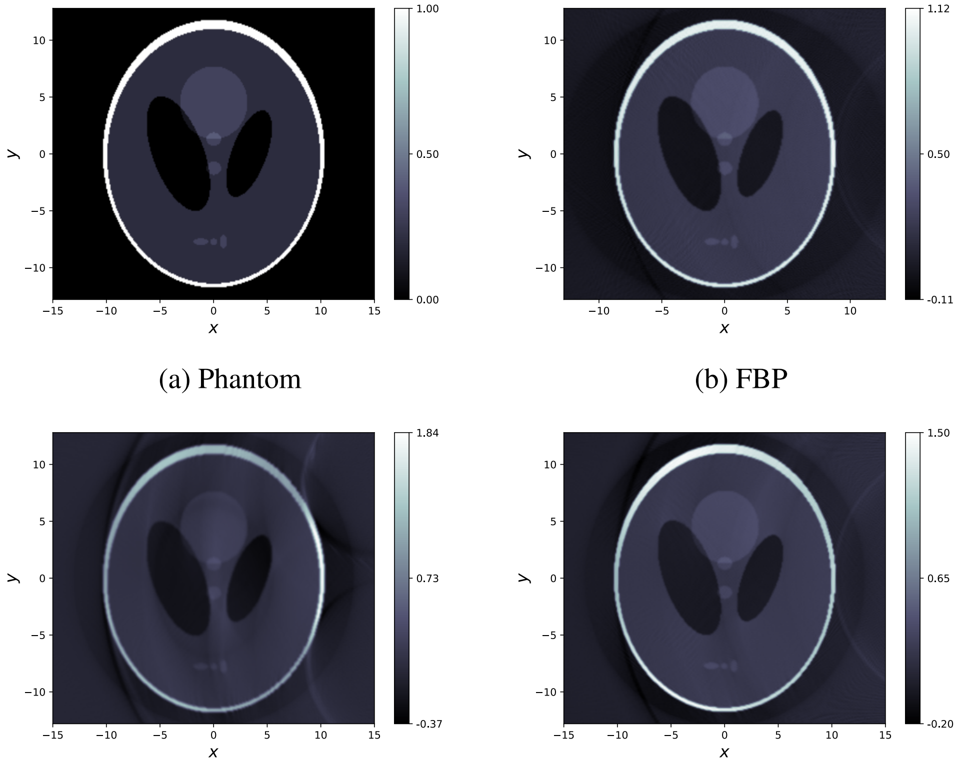
(c) Katsevich (K)

(d) Katsevich (NPH)

Figure 5.1: Shepp-Logan reconstruction with different methods using flat detector and 600 source positions.

$[0, 1]$ range, the reconstructions include slightly negative values. This is not particularly problematic, as we are mainly interested in relative contrast rather than the absolute intensity of the voxels.

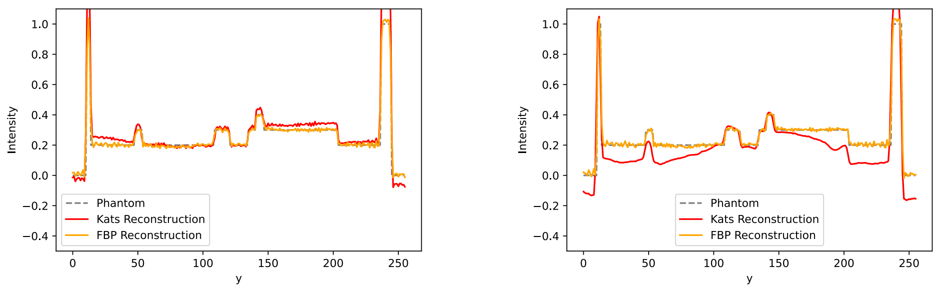
Upon visual inspection, the reconstructions appear overall similar. Katsevich methods show slightly more blurring in the flat regions of the phantom, but they manage to preserve the overall shape of the phantom. The metrics in Table 5.1 confirm these observations: PSNR favors FBP due to voxel-wise accuracy, while SSIM indicates that Katsevich retains better the structural information.



(a) Phantom (b) FBP

(c) Katsevich (K) (d) Katsevich (NPH)

Figure 5.2: Shepp-Logan reconstruction using curved detector and 600 source positions.



(a) Flat (b) Curved

Figure 5.3: Attenuation profile of the line $x = 0$ and $z = 0$ (middle line) for Katsevich and FBP method.

For the curved detector configuration, the results are notably worse. Looking at Figure 5.2, we notice that both Katsevich variants exhibit stronger artifacts within the phantom, especially in regions of uniform intensity. This is evident in the attenuation profile shown in Figure 5.3, where Katsevich tends to dip in flat areas, unlike FBP, which follows the true profile of the phantom more closely. This effect only appears in the curved detector case, suggesting that the cause may be an inconsistency in mapping the detector data to the reconstruction volume. As confirmed by Table 5.1, Katsevich performs worse than FBP in PSNR, although SSIM remains comparable, indicating that the overall structure is still largely preserved.

Views	Detector	Method	PSNR [dB]	SSIM
60	Flat	FBP	16.93	0.343
60	Flat	Kats K	15.46	0.370
60	Flat	Kats NPH	15.97	0.386
60	Curved	FBP	16.94	0.341
60	Curved	Kats K	15.25	0.343
60	Curved	Kats NPH	15.54	0.344
300	Flat	FBP	23.88	0.510
300	Flat	Kats K	21.28	0.530
300	Flat	Kats NPH	20.72	0.542
300	Curved	FBP	23.83	0.507
300	Curved	Kats K	19.07	0.486
300	Curved	Kats NPH	18.94	0.488
600	Flat	FBP	24.56	0.581
600	Flat	Kats K	21.32	0.584
600	Flat	Kats NPH	20.99	0.589
600	Curved	FBP	24.13	0.545
600	Curved	Kats K	19.10	0.500
600	Curved	Kats NPH	19.08	0.500

Table 5.1: Quantitative results for the Shepp-Logan phantom under different angular sampling and detector configurations.

We next analyze how the methods behave under different levels of angular sparsity. As seen in Table 5.1, Katsevich generally demonstrates superior ability in preserving the structural content of the phantom, as indicated by consistently higher SSIM values compared to FBP. On the other hand, FBP tends to achieve higher PSNR across all configurations, indicating better preservation of absolute voxel intensities. As expected, the values of the metrics improve with increasing numbers of views, and all methods tend to stabilize around 600 views. This trend is clearly illustrated in Figure 5.4, which shows the evolution of PSNR and SSIM for each method across different levels of angular sampling.

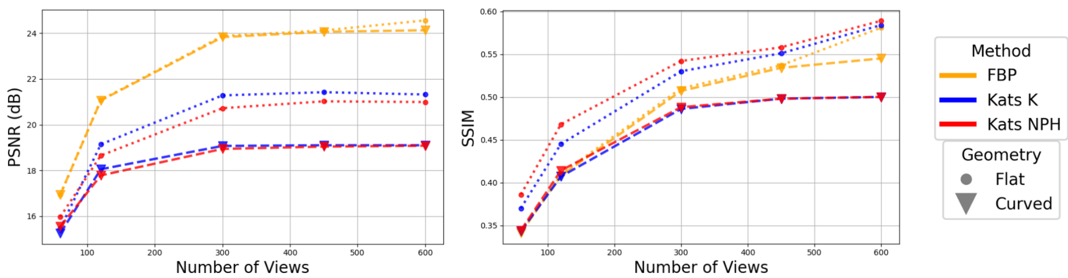
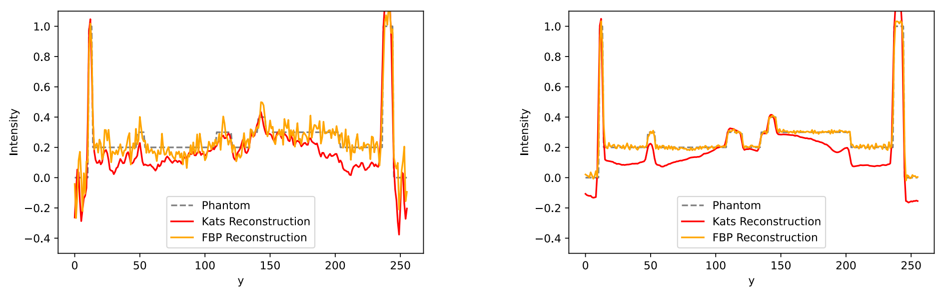


Figure 5.4: Quantitative metrics of FBP and Katsevich methods using different angular sparsity.

In the sparse sampling case (*e.g.,* 60 views), both flat and curved detector configurations show a clear advantage for Katsevich in terms of SSIM, with PSNR values that are close to those of FBP. As the number of views increases (*e.g.,* to 300), the two detector configurations begin to show diverging trends. In the flat case, Katsevich continues to outperform FBP in SSIM, although the gap narrows, while FBP remains superior in PSNR. On the other hand, in the curved configuration FBP surpasses Katsevich in both SSIM and PSNR as the sampling increases.



(a) 60 views

(b) 300 views

Figure 5.5: Attenuation Profile of the line $x = 0$ and $z = 0$ (middle line) for Katsevich and FBP method.

This performance degradation in the curved case can be attributed to the systematic drop in reconstructed intensity in flat regions of the phantom when using Katsevich. While this effect is less pronounced under sparse sampling where the undersampling error dominates, it becomes more significant as the number of source positions increases, see Figure 5.5. This intensity error may be influenced by numerical instabilities in the weighting scheme used during backprojection or by the high sensitivity of Katsevich to the acquisition parameters. The geometry settings adopted here (see Table 4.1) follow those suggested in another Katsevich implementation [17], but they may not be optimal for this specific configuration.

Interestingly, although the literature often reports better performance for the Katsevich K variant [28], our experiments show that the two derivative schemes yield very similar results. Remarkably, the NPH scheme tends to preserve structural information slightly better, achieving higher SSIM scores across both detector configurations as can be seen in Figure 5.4. Given the overall comparability of the two approaches, the following sections will report results using only the Katsevich method with the K derivative scheme.

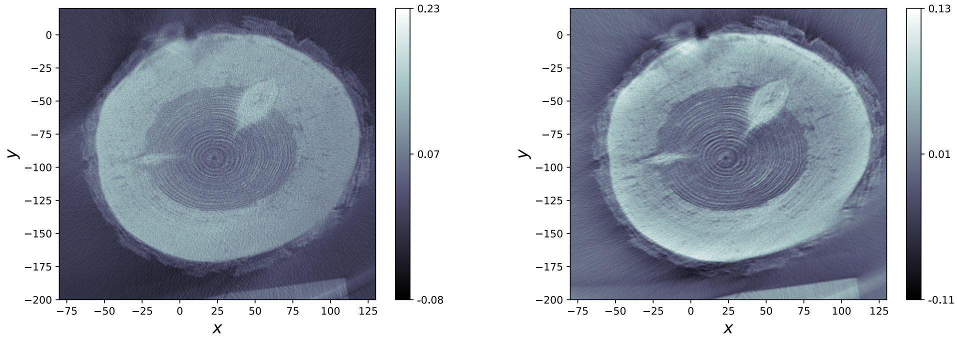
The code used to perform these experiments, along with additional results, is available in the repository at odl-katsevich under the directory `odl/examples/kats/`.

5.2 Wooden Logs Data

We now evaluate the reconstruction performance on the wooden log dataset, using the standard FBP method as reference since no ground truth is available.

Figure 5.6 shows the reconstruction of the central slice, where the internal structure and nodules are clearly visible. The FBP reconstruction exhibits mild streak artifacts. In contrast, the Katsevich reconstruction displays noticeably stronger artifacts and intensity fluctuations in regions expected to be homogeneous, particularly in sections of the wood interior close to and adjacent to the bark, where there is a significant intensity jump. However, the Katsevich method appears to produce a sharper image, enhancing contrast and capturing more effectively the imperfections in the internal structure of the wood.

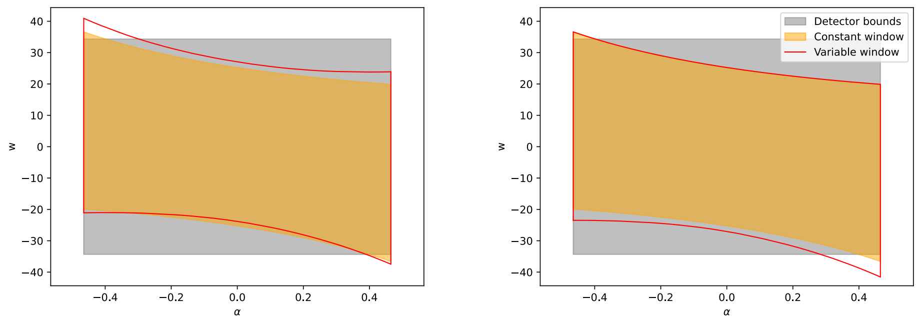
One of the major differences from the synthetic experiments is the variable pitch used in the acquisition. This change has a substantial impact on



(a) FBP Reconstruction. (b) Katsevich Reconstruction.

Figure 5.6: Slice of the reconstruction of a wooden log at $z = 184.96$ mm.

computational complexity, especially during the initialization of the filter and in the application of the TD window. Since the pitch varies across views, both the forward and backward rebinning maps must be recomputed individually for each view; as a result, the pre-processing steps become significantly more time-consuming. The same applies to the TD window masks; more details about this can be found in Appendix D.2. Figure 5.7 illustrates a comparison between the TD window computed under the assumption of constant pitch and that adapted for variable pitch, for different source positions s.



(a) View 2332 of 5264. (b) View 3935 of 5264.

Figure 5.7: Comparison of the variable TD windows for different source positions (in red) and the TD window resulting from using constant pitch (in yellow). The grey area represents the detector plate.

To quantify the impact of variable pitch on computational complexity, Table 5.2 reports the runtime for the key components of the reconstruction pipelines: filter initialization, data filtering, TD window computation, and

Method	Filter Init. [s]	Filtering [s]	TD Window [s]	BP [s]	**Total [s]**
FBP	7.28	291.20	–	23.67	**333.02**
Katsevich (variable pitch)	6943.00	534.04	53.54	23.79	**7554.37**
Katsevich (constant pitch)	56.97	509.91	3.63	23.88	**594.39**

Table 5.2: Runtime per stage for FBP and Katsevich reconstructions.

backprojection. The comparison includes FBP, the Katsevich method assuming constant pitch (i.e., ignoring the pitch variations), and the full Katsevich method that recomputes the rebinning mappings and the TD window for each view to account for pitch variations.

The results show that the filter initialization step, which is relatively fast for FBP and for Katsevich with constant pitch, becomes the dominant bottleneck in the variable-pitch Katsevich method, requiring approximately 100 minutes. The remaining stages have comparable runtimes across all methods. Note that in FBP, the windowing and backprojection steps are integrated, so they are not reported separately.

Given that the reconstruction of a single slice using the full Katsevich method (with variable pitch) took around 125 minutes, reconstructing the entire volume of 738 slices would take an impractically long time. However, it is important to note that the reconstruction of a single slice, as shown in Figure 4.1, was performed using 3386 out of the total 11066 projections (so roughly one third of the total).

Therefore, if the current memory limitations that restrict the reconstruction to one slice at a time can be overcome, it would be possible to reconstruct the entire volume in a single pass. In that case, the total reconstruction time would scale roughly with the number of projections considered, resulting in an expected runtime approximately three times longer than that for a single slice (i.e., 11066/3386).

An open question regards the actual influence of the TD window variation on reconstruction quality. It would be worthwhile to investigate whether these the recomputation of a window for each source position, which select different subsets of data on the detector for reconstruction, result in meaningful improvements or could be neglected for the sake of a much faster reconstruction. Clearly, this assessment must be carried out on a case-by-case basis, as the impact strongly depends on the extent of pitch variation. Nevertheless, further study is recommended.

5.3 Clinical Data

The discussion of the results of applying the Katsevich reconstruction pipeline on the Mayo dataset now focuses on chest and abdomen scans acquired with a full dose regime. Note that the code used to perform these experiments, along with additional results, is available in the repository at odl-katsevich under the directory `odl/odl/contrib/datasets/ct/examples/`.

Figure 5.8 shows reconstructions of the chest scan C012 using both FBP and Katsevich methods; the corresponding acquisition geometry parameters are shown in Table 4.3. Additional reconstructions of other scans of the dataset are included in Appendix E.

In the axial view, the Katsevich reconstruction shows more streaking and the typical discoloration in flat regions already noted and discussed in previous sections of the results. However, in the sagittal and coronal views, the standard FBP reconstruction suffers from severe streak artifacts. This is likely due to FBP not being designed for the double-helix geometry introduced by FFS. In contrast, Katsevich reconstructions are more robust to this geometry and show significantly fewer artifacts, even though our TD windowing is not fully adapted to the FFS and double-helix model.

The sharpness of the Katsevich results varies across scans. Some reconstructions are very sharp, while others appear slightly blurred. For example, the reconstruction of scan C012 is noticeably sharper than that of scan C004, as shown in Figure 5.9. Interestingly, it does not seem to depend on the number of projections used for the reconstruction, as the C004 scan is obtained using a bigger amount of projections than the C012 scan (13782 against 13446).

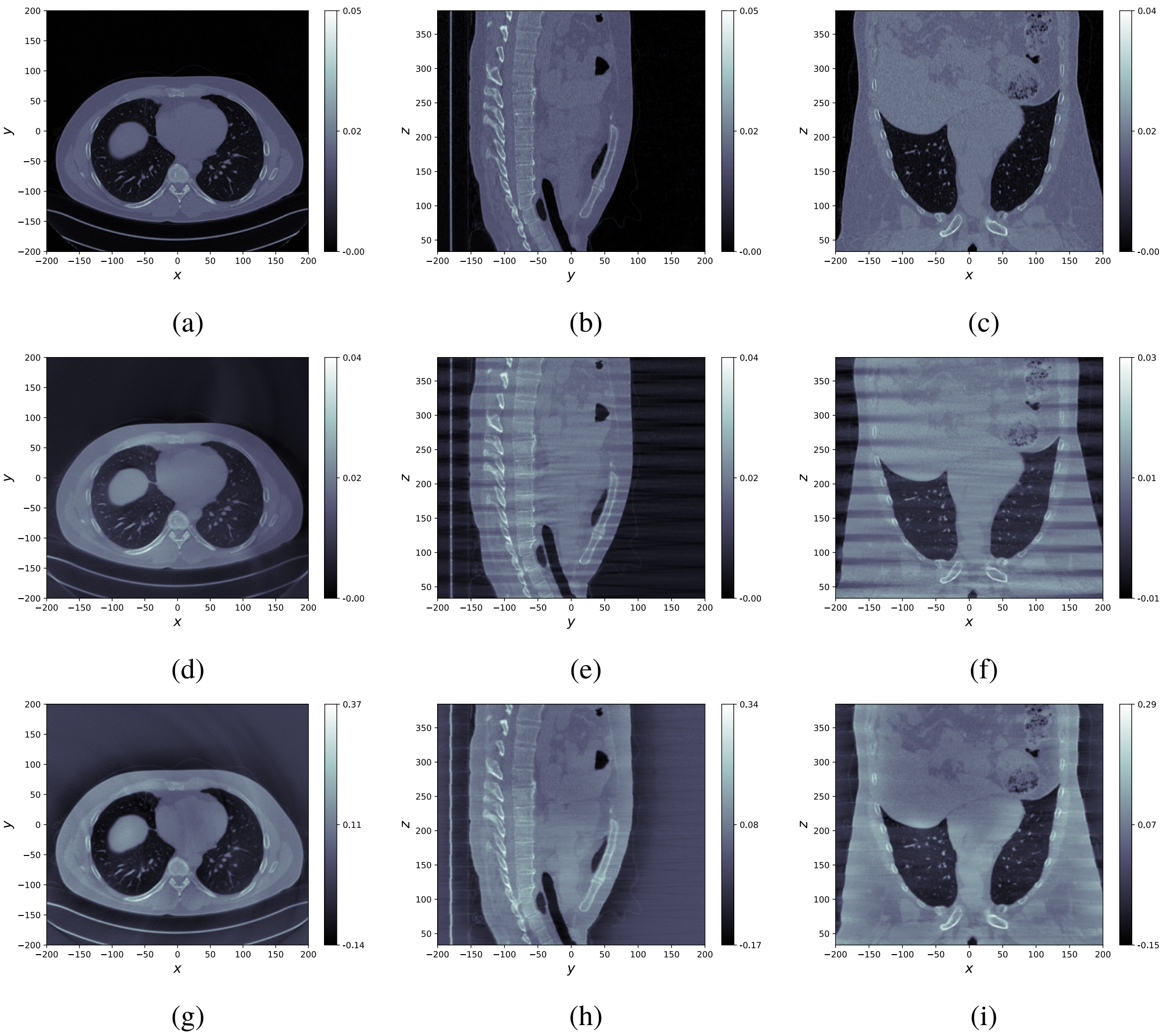
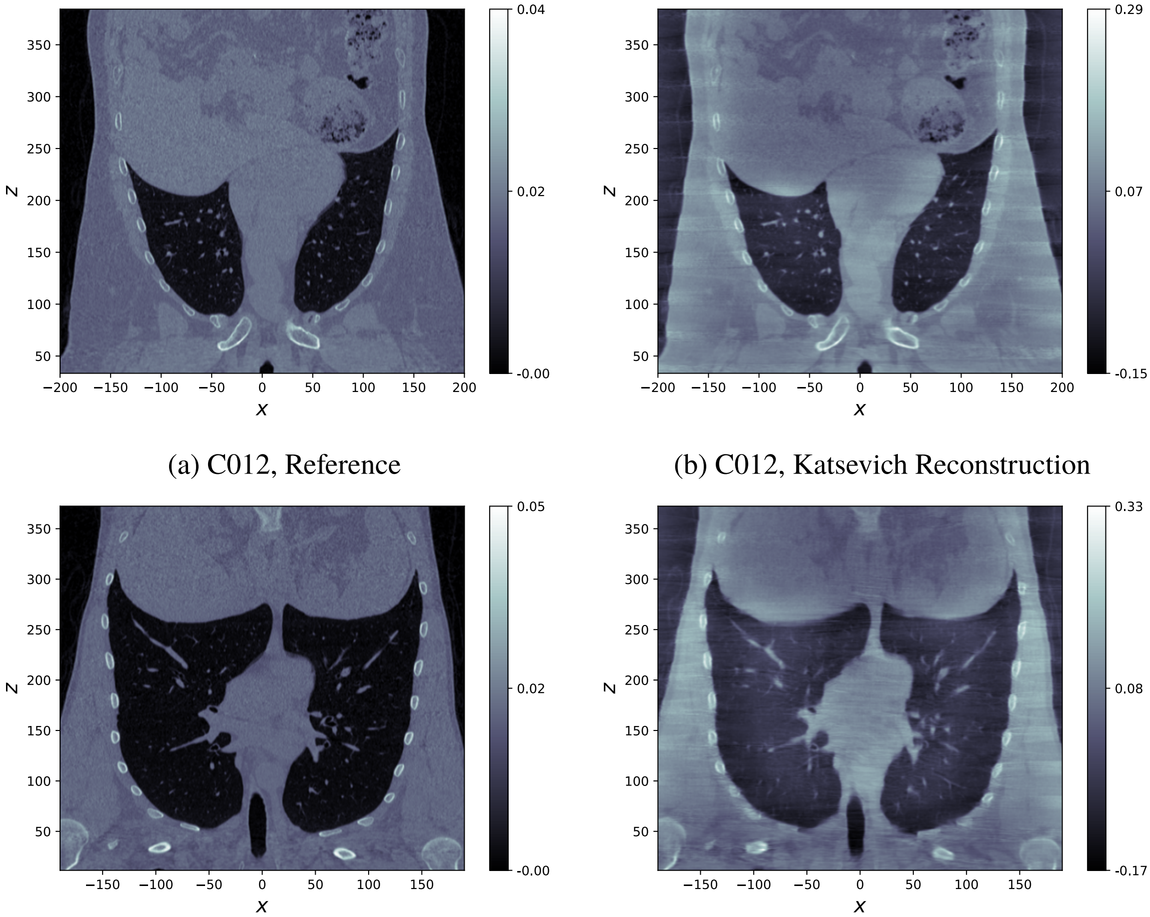


Figure 5.8: Axial, sagittal and coronal views of the reconstructions of the chest scan C012 in the Mayo dataset using FBP 5.9d, 5.8e, 5.8f and Katsevich 5.8g, 5.8h, 5.8i. The reference reconstruction 5.9a, 5.9b, 5.9c is provided by the Mayo dataset and is executed using an FBP-type method. The acquisition geometry parameters are reported in Table 4.3. Obtained from a sinogram of 13446 projections.



(a) C012, Reference

(b) C012, Katsevich Reconstruction

(c) C004, Reference

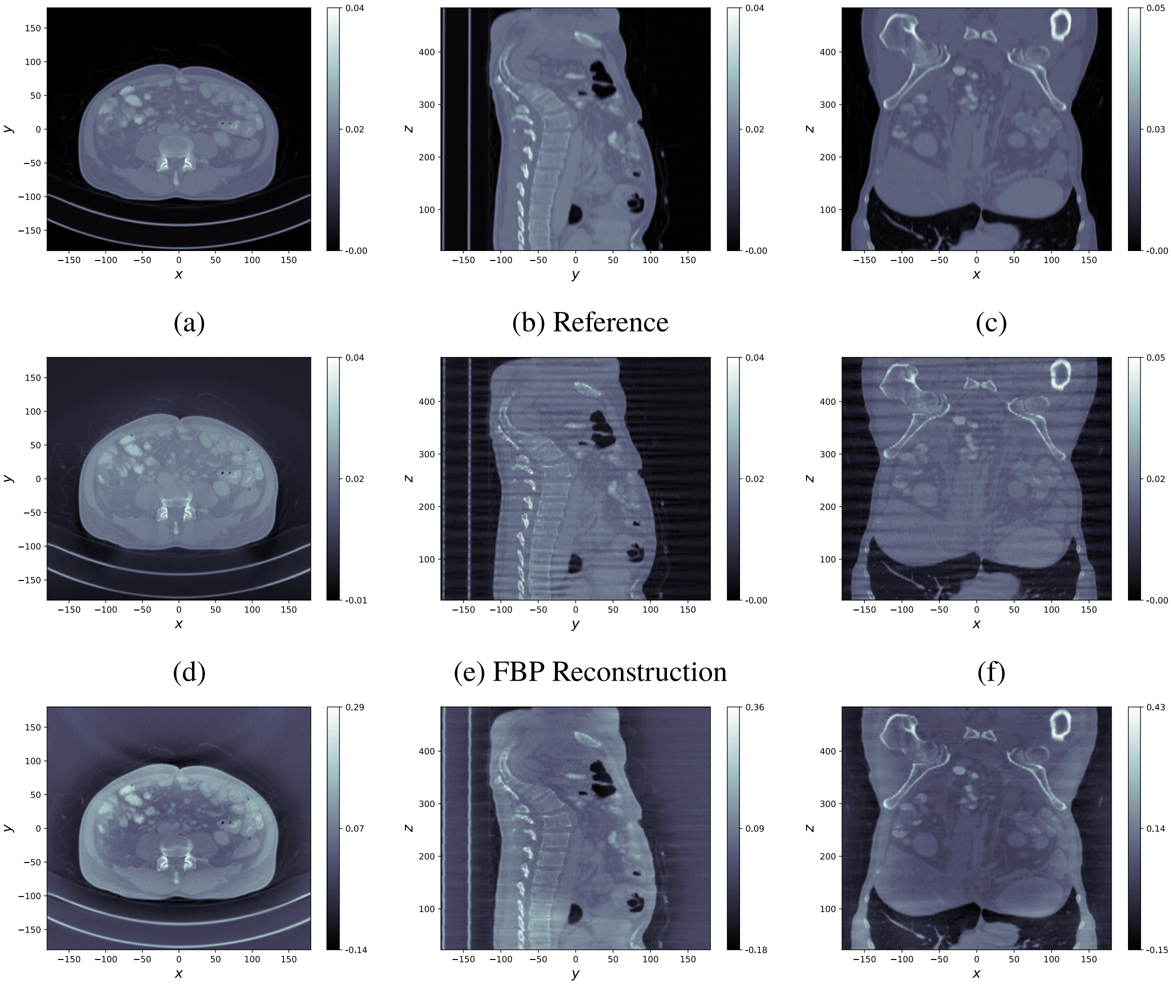
(d) C004, Katsevich Reconstruction

Figure 5.9: Comparison of the quality of the image in the Katsevich reconstruction of the two scans C012 and C004. The reconstruction of C004 is clearly more blurred out than the reconstruction of C012.

Similar trends are seen in reconstructing abdominal scans, including the one shown in Figure 5.10. The overall results confirm that while Katsevich may introduce some blurring in certain cases, it better handles the complexities introduced by the helical trajectory and the flying focal spot geometry, especially in reconstructing slices of the volume in the rotation axis direction (z).

The quantitative metrics still favor the FBP method. While the higher PSNR for FBP is expected, the SSIM results are somewhat surprising.

Several factors could explain this result. First, the reference image used as ground truth is reported to have been obtained with an FBP-type method,



(a) (b) Reference (c)

(d) (e) FBP Reconstruction (f)

(g) (h) Katsevich Reconstruction (i)

Figure 5.10: Axial, sagittal and coronal views of the reference images 5.10a, 5.10b, 5.10c and the reconstructions of the abdomen scan L014 in the Mayo dataset using FBP 5.10d, 5.10e, 5.10f and Katsevich 5.10g, 5.10h, 5.10i. Obtained from a sinogram of 37980 projections. The parameters of the acquisition geometry are shown in Table 4.3.

Dataset	Method	PSNR [dB]	SSIM
C012 (Chest)	FBP	24.59	0.517
	Katsevich	12.38	0.330
C004 (Chest)	FBP	23.33	0.528
	Katsevich	13.64	0.337
L014 (Abdomen)	FBP	26.16	0.586
	Katsevich	13.99	0.395

Table 5.3: Quantitative results for FBP and Katsevich reconstructions (rescaled to the range of the reference image) across different datasets.

which may inherently favor similar reconstruction approaches in the evaluation of such quantitative metrics. Additionally, the Katsevich reconstruction tends to span a broader range of intensity values compared to FBP. This mismatch in scale likely penalizes the Katsevich results when metrics such as SSIM and PSNR are computed.

Dataset	FBP [s]	Katsevich [s]
C012	216.92	551.72
C004	210.47	568.94
L014	587.32	1580.26

Table 5.4: Total reconstruction time.

Dataset	Filter Init. [s]	Filtering [s]	TD+BP [s]
C012	3.09	253.00	28.46
C004	2.93	535.75	30.26
L014	3.06	1502.11	75.09

Table 5.5: Runtime breakdown for Katsevich.

The runtime results in Table 5.4 and Table 5.5 align with those reported in Section 5.2. Having a constant pitch, not considering the oscillations introduced by FFS, the most time-consuming part of the Katsevich pipeline becomes the filtering step. As expected, its computational cost scales with the number of projections in the dataset. In general, the total runtime of the reconstruction using Katsevich is roughly 2-3 times the one of FBP.

Chapter 6

Discussion

The results obtained with the implemented Katsevich reconstruction pipeline are promising: the method consistently preserves the structural information of the object, handles helical trajectories, and produces reconstructions that are often sharper or less affected by helical artifacts than standard FBP. However, it is equally clear that Katsevich, in its current implementation, falls short of expectations. In many cases, especially with real data, FBP outperforms it both quantitatively and qualitatively. This discrepancy likely stems from a combination of technical issues affecting stages of the pipeline.

First of all, the implementation faced significant constraints, particularly in the backprojection step. Since the ASTRA Toolbox does not support voxel- and view-dependent weighting during backprojection, it was not possible to directly encode the Katsevich formula using the standard `ray_trafo.adjoint` operator in ODL. A custom workaround was developed via a new method (`adjoint_kats`) that applies ASTRA's adjoint operator followed by voxel-wise scaling using *CuPy*. While this allows the method to approximate the correct behavior, it introduces two limitations: it breaks the modular operator interface of ODL, and it relies on an external backend *CuPy* not officially integrated in the ODL package. A more elegant alternative, decomposing the backprojection into per-view operators, was outlined but not fully implemented in this work.

Beyond implementation constraints, a further issue arose when evaluating reconstructions that involve curved detectors. While reconstructions with flat detectors show good performance and closely match theoretical expectations, the curved detector case reveals some deviations. In flat regions of the phantom, the Katsevich reconstruction systematically overestimates attenuation, resulting in visible dips in the intensity profile. These artifacts are not present after filtering and appear only after backprojection, suggesting that they are caused by a wrong or inaccurate application of the weighting in curved geometries. The fact that FBP does not exhibit the same problem supports the hypothesis of the involvement of the weights.

Another layer of complexity is added when extending the method to clinical datasets, where the acquisition geometry includes variable pitch and flying focal spot (FFS). While the implementation successfully handles variations in pitch by recomputing filtering and TD windows per view, the effects of FFS are only partially addressed. In particular, only offsets in the rotation axis (z-axis) are fully modeled, while lateral shifts of the focal spot, affecting both the trajectory and the source-to-detector distance D, are only partially incorporated into the filtering step, *i.e.,* the source position being explicitly stored is correctly handled, but the variation in source-to-detector distance is not accounted for in the filtering computation. The current implementation also applies the TD window approximately, without creating separate windows for the two helical paths. Despite these simplifications, Katsevich still manages to suppress artifacts in sagittal and coronal planes more effectively than FBP, highlighting the method is naturally fit to helical sampling. This contrast is particularly visible in the Mayo Clinic datasets, where FBP suffers from pronounced streaks aligned with the rotation axis, while Katsevich reduces them to an acceptable level.

Quantitatively, the situation is mixed. In synthetic settings and under angular undersampling, Katsevich often achieves higher SSIM scores, confirming better structural preservation, while FBP tends to produce higher PSNR, indicating better voxel-wise accuracy. On clinical data, where no ground truth is available, numerical metrics still favor FBP, possibly due to a bias introduced by using FBP-based reference reconstructions, but visual inspection suggests that Katsevich provides cleaner, more artifact-free reconstructions in certain orientations, which may have practical relevance in diagnostic contexts.

Runtime is another critical consideration. While the filtering and backprojection stages are reasonably efficient, the pre-processing cost in the variable-pitch case, especially during filter initialization, is extremely high due to the need to recompute rebinning maps and TD windows for each view. In the worst case, where the resolution of the detector is very high, a single slice reconstruction can require hours, rendering the full-volume reconstruction impractical without further optimization.

In conclusion, while the Katsevich method as implemented shows promise and clearly benefits from its theoretical suitability for helical CT, its current limitations, stemming from both geometric modeling and backend constraints, prevent it from outperforming standard FBP in most practical scenarios. Nonetheless, its ability to reduce motion-induced artifacts and preserve anatomical structure, especially in challenging geometries, points toward valuable future directions. Improvements in the curved detector modeling, full support for FFS, and a better integration of voxel-weighted backprojection into ODL could allow the Katsevich method to fully realize its potential as a high-fidelity alternative to standard FBP-type methods.

Chapter 7

Conclusions

This thesis was set out to implement a flexible and modular version of the Katsevich reconstruction algorithm for 3D helical CT, supporting realistic acquisition settings such as curved detectors, variable pitch, and flying focal spot (FFS). The method was successfully integrated into the ODL framework and validated on synthetic, experimental, and clinical data.

While the algorithm consistently preserves structural information and shows robustness to helical sampling, especially in sagittal and coronal views, it does not yet outperform FBP across all scenarios. Several limitations remain, primarily due to the constraints imposed by the ASTRA backend, which prevents direct application of voxel- and view-dependent weights during backprojection. Inaccuracies also emerge in the curved detector case, and FFS effects are only partially accounted for in the reconstruction. Moreover, the computational cost, particularly under variable pitch, limits the method's practicality for the reconstruction of full volumes, limiting the reconstruction to a slice by slice approach, when the detector presents a very high resolution. Nonetheless, these challenges have clarified where the implementation of the method can be improved and point to clear directions for future work.

A priority would be to develop a ray-by-ray backprojection routine or extend ASTRA's capabilities to support voxel-wise weighting directly. A more complete and accurate handling of FFS, especially in the filtering and windowing steps, is also essential. Runtime could be improved by exploiting sparsity in the weighting and filtering operations; for instance, methods like that proposed by Qiao et al. [12] suggest skipping certain steps in the filtering stage when the cone angle is small.

Beyond algorithmic improvements, future work could include validation on low-dose datasets to assess robustness to noise, and comparison to more assessed methods like iterative methods.

Overall, this thesis contributes a working, open-source Katsevich implementation and demonstrates its potential in settings where conventional FBP struggles. While some objectives remain partially met, the work establishes a solid foundation for further research and refinement in exact analytical reconstruction methods in 3D computed tomography.

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Appendix A
K-lines formula for constant pitch

We want to evaluate the expression

$$w(u) = -\frac{D\,\mathbf{e}_v \cdot \mathbf{n} + u\,\mathbf{e}_u \cdot \mathbf{n}}{\mathbf{e}_w \cdot \mathbf{n}}, \tag{A.1}$$

with

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2, \quad \mathbf{v}_1 = y(s+\psi) - y(s), \quad \mathbf{v}_2 = y(s+2\psi) - y(s), \tag{A.2}$$

and the parametric helix

$$y(s) = \begin{bmatrix} R\cos s \\ R\sin s \\ \frac{P}{2\pi}s \end{bmatrix}, \qquad \psi(s) = \frac{P}{2\pi}s. \tag{A.3}$$

We compute

$$\mathbf{v}_1 = \begin{bmatrix} R(\cos(s+\psi) - \cos s) \\ R(\sin(s+\psi) - \sin s) \\ \frac{P}{2\pi}\psi \end{bmatrix}, \tag{A.4}$$

$$\mathbf{v}_2 = \begin{bmatrix} R(\cos(s+2\psi) - \cos s) \\ R(\sin(s+2\psi) - \sin s) \\ \frac{P}{2\pi}(2\psi) \end{bmatrix}. \tag{A.5}$$

The cross product $\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2$ has components:

$$n_x = R(\sin(s+\psi) - \sin s) \cdot \frac{P}{2\pi}(2\psi) - R(\sin(s+2\psi) - \sin s) \cdot \frac{P}{2\pi}\psi, \tag{A.6}$$

$$n_y = -R(\cos(s+\psi) - \cos s) \cdot \frac{P}{2\pi}(2\psi) + R(\cos(s+2\psi) - \cos s) \cdot \frac{P}{2\pi}\psi, \tag{A.7}$$

$$n_z = -R^2(\sin(s+\psi) - \sin s)(\cos(s+2\psi) - \cos s) + R^2(\sin(s+2\psi) - \sin s)(\cos(s+\psi) - \cos s). \tag{A.8}$$

The basis vectors are:

$$\mathbf{e}_u = (-\sin s, \cos s, 0), \qquad \mathbf{e}_v = (-\cos s, -\sin s, 0), \qquad \mathbf{e}_w = (0, 0, 1). \tag{A.9}$$

Then the scalar products are:

$$\mathbf{e}_v \cdot \mathbf{n} = -\cos s \cdot n_x - \sin s \cdot n_y, \tag{A.10}$$

$$\mathbf{e}_u \cdot \mathbf{n} = -\sin s \cdot n_x + \cos s \cdot n_y, \tag{A.11}$$

$$\mathbf{e}_w \cdot \mathbf{n} = n_z. \tag{A.12}$$

The numerator of $w(u)$ becomes:

$$D\,\mathbf{e}_v \cdot \mathbf{n} + u\,\mathbf{e}_u \cdot \mathbf{n} = -D(\cos s\, n_x + \sin s\, n_y) - u(\sin s\, n_x - \cos s\, n_y). \tag{A.13}$$

Putting it all together:

$$w(u) = \frac{P\psi\left[-D(2\sin\psi - \sin(2\psi)) - u(2\cos\psi - \cos(2\psi) - 1)\right]}{2\pi R(-2\sin\psi + \sin(2\psi))}. \tag{A.14}$$

Rearranging,

$$w(u) = \frac{DP}{2\pi R}\left(\frac{\psi(-2\sin\psi + \sin(2\psi))}{-2\sin\psi + \sin(2\psi)} + \frac{u\psi(\cos(2\psi) + 1 - 2\cos(\psi))}{D(-2\sin(\psi) + \sin(2\psi))}\right) \tag{A.15}$$

$$= \frac{DP}{2\pi R}\left(\psi + \frac{u\psi(2\cos(\psi)(\cos(\psi) - 1))}{D(2\sin(\psi)(\cos(\psi) - 1))}\right) \tag{A.16}$$

$$= \frac{DP}{2\pi R}\left(\psi + \frac{u\psi}{D\tan(\psi)}\right). \tag{A.17}$$

Appendix B

Proofs for Katsevich in local detector coordinates

This section gives a more in depth explanation to how to express the various terms of the Katsevich inversion formula in local coordinates

B.1 Flat detector

Derivative term. Proof of Equation (2.39).

Lemma 3. *Let:*

$$g_1(s,u,w) = \frac{\partial}{\partial q} g(q,u,w)\Big|_{q=s} \tag{B.1}$$

Then:

$$g_1(s,u,w) = \left(\frac{\partial g}{\partial q} + \frac{u^2 + D^2}{D} \frac{\partial g}{\partial u} + \frac{uw}{D} \frac{\partial g}{\partial w} \right)\Big|_{q=s} \tag{B.2}$$

Proof. We differentiate $g(q,u,w)$ with respect to q using the chain rule:

$$\frac{\partial}{\partial q} g_f(q,u,w) = \frac{\partial g_f}{\partial q} + \frac{\partial g_f}{\partial u}\frac{\partial u}{\partial q} + \frac{\partial g_f}{\partial w}\frac{\partial w}{\partial q}. \tag{B.3}$$

Given the ray direction:

$$\theta_f(s,u,w) = \frac{1}{\sqrt{u^2 + D^2 + w^2}} \left(u\,\mathbf{e}_u(s) + D\,\mathbf{e}_v(s) + w\,\mathbf{e}_w \right),$$

we express the coordinates of the intersection with the detector as:

$$u = D\frac{\langle \theta_f, \mathbf{e}_u(q)\rangle}{\langle \theta_f, \mathbf{e}_v(q)\rangle}, \quad w = D\frac{\langle \theta_f, \mathbf{e}_w\rangle}{\langle \theta_f, \mathbf{e}_v(q)\rangle}.$$

Using $\mathbf{e}'_u = \mathbf{e}_v$, $\mathbf{e}'_v = -\mathbf{e}_u$, and $\mathbf{e}'_w = 0$, we find:

$$\begin{aligned} \frac{\partial u}{\partial q} &= \frac{D(\langle \theta_f, \mathbf{e}_v\rangle^2 + \langle \theta_f, \mathbf{e}_u\rangle^2)}{\langle \theta_f, \mathbf{e}_v\rangle^2} = \frac{u^2 + D^2}{D}, \\ \frac{\partial w}{\partial q} &= \frac{D\langle \theta_f, \mathbf{e}_w\rangle\langle \theta_f, \mathbf{e}_u\rangle}{\langle \theta_f, \mathbf{e}_v\rangle^2} = \frac{uw}{D}. \end{aligned}$$

Substituting back yields the result. □

Convolution term. Proof of Equation (2.44).

Lemma 4. *Let $(u, w_\kappa(u))$ be the intersection point of detector plane $DP(s)$ and a ray $\theta(s, x, \gamma) = \cos\gamma\beta(s, x) + \sin\gamma V(s, x)$ on the κ−plane defined by β and e, and $(u', w'_\kappa(u'))$ the intersection point of detector plane $DP(s)$ and the ray $\beta(s, x)$ that originates from $y(s)$ and passes through x. Let $r = \sqrt{u^2 + D^2 + w_\kappa^2(u)}$ and $r' = \sqrt{(u')^2 + D^2 + (w'_\kappa(u'))^2}$ be the distances from $y(s)$ to the detector points $(u, w_\kappa(u))$ and $(u', w'_\kappa(u'))$, respectively. Then the convolution 2.31*

$$\int_0^{2\pi} k_{\mathrm{H}}(\sin(\gamma)) g'(s, \theta(s, x, \gamma))\, d\gamma \tag{B.4}$$

can be written as:

$$g^F(s, \beta) = -\frac{r}{D}\int_{-\infty}^{\infty} k_H(u - u')\frac{D}{r'} g_1(s, u', w'_\kappa(u'))du'. \tag{B.5}$$

Proof. □

Integrand. Proof of Equation (2.45).

Lemma 5. *Let*

$$\begin{aligned} g^F(s, u, w_\kappa(u)) &:= \int_{-\infty}^{\infty} k_H(u - u')\frac{D g_1(s, u', w'_\kappa(u'))}{\sqrt{(u')^2 + D^2 + (w'_\kappa(u'))^2}} du' \\ &= -\frac{D}{r} g^F(s, \beta). \end{aligned} \tag{B.6}$$

Then:

$$\frac{-1}{|x - y(s)\|} g^F(s, \beta) = \frac{1}{R + \langle x, \mathbf{e}_v \rangle} g^F(s, u, w_\kappa(u)). \tag{B.7}$$

Proof. (necessary to include it?) From Theorem 4 we have:

$$g^F(s, \beta) = -\frac{r}{D} g^F(s, u, w_\kappa),$$

with $r = \sqrt{u^2 + D^2 + w_\kappa^2}$.

By similar triangles in the projection geometry:

$$\frac{r}{D} = \frac{\|x - y(s)\|}{\langle x - y(s), \mathbf{e}_v \rangle} = \frac{\|x - y(s)\|}{R + \langle x, \mathbf{e}_v(s) \rangle}.$$

Therefore:

$$-\frac{1}{\|x - y(s)\|} g^F(s, \beta) = \frac{1}{\|x - y(s)\|} \frac{r}{D} g^F(s, u, w_\kappa) = \frac{1}{v^*(s, x)} g^F(s, u, w_\kappa),$$

as claimed. □

B.2 Curved detector

Derivative term. Proof of Equation (2.51).

Lemma 6. *Let:*

$$g_1(s, \alpha, w) = \frac{\partial}{\partial q} g(q, \alpha, w)\Big|_{q=s} \tag{B.8}$$

Then:

$$g_1(s, u, w) = \left(\frac{\partial g}{\partial q} + \frac{\partial g}{\partial \alpha} \right)\Big|_{q=s} \tag{B.9}$$

Proof. Again using the chain rule we get:

$$\frac{\partial}{\partial q} g(q, \alpha, w) = \frac{\partial g}{\partial q} + \frac{\partial g}{\partial \alpha}\frac{\partial \alpha}{\partial q} + \frac{\partial g}{\partial w}\frac{\partial w}{\partial q}. \tag{B.10}$$

Using the expressions in 2.49 for the intersection (α, w) between a ray θ_c and the detector, we obtain:

$$\frac{\partial w}{\partial q} = 0,$$

since w is independent of s, and

$$\begin{aligned}
\frac{\partial \alpha}{\partial q} &= \frac{\left(\frac{\langle \theta, \mathbf{e}_u(s)\rangle}{\langle \theta, \mathbf{e}_v(s)\rangle}\right)'}{1 + \left(\frac{\langle \theta, \mathbf{e}_u(s)\rangle}{\langle \theta, \mathbf{e}_v(s)\rangle}\right)^2} \\
&= \frac{\left(\langle \theta_c, \mathbf{e}_v(s)\rangle^2 + \langle \theta_c, \mathbf{e}_u(s)\rangle^2\right)}{\left(1 + \left(\frac{\langle \theta, \mathbf{e}_u(s)\rangle}{\langle \theta, \mathbf{e}_v(s)\rangle}\right)^2\right) \langle \theta_c, \mathbf{e}_v(s)\rangle^2} \\
&= \frac{1}{1 + \tan^2 \alpha} \frac{1}{\cos^2 \alpha} \\
&= 1.
\end{aligned}$$

Substituting back yields the result. $\square$

Convolution term. Proof of Equation (2.54).

Lemma 7. *The result of Theorem 4 can be extended to the curved case by:*

$$g^F(s, \beta) = -\frac{\sqrt{D^2 + w_\kappa(\alpha)^2}}{D} \int_{-\pi/2}^{\pi/2} k_H(\sin(\alpha - \alpha')) \frac{Dg_1(s, \alpha', w'_\kappa(\alpha'))}{\sqrt{D^2 + w'_\kappa(\alpha')^2}} d\alpha'. \tag{B.11}$$

Proof. The above mentioned formula is obtained simply substituting Equation (2.50) for u and w in the result obtained in Theorem 4. $\square$

Integrand. Proof of Equation (2.55).

Lemma 8. *Let*

$$g^F(s, \alpha, w_\kappa) := \cos \alpha \int_{-\pi/2}^{\pi/2} k_H(\sin(\alpha - \alpha')) \frac{Dg_1(s, \alpha', w'_\kappa)}{\sqrt{D^2 + w'_\kappa(\alpha')^2}} d\alpha'$$

Then:

$$\frac{-1}{|x - y(s)\|} g^F(s, \beta) = \frac{1}{R + \langle x, \mathbf{e}_v \rangle} g^F(s, \alpha, w_\kappa(u)). \tag{B.12}$$

Proof. From Theorem 7 we have:

$$g^F(s,\beta) = -\frac{r}{D} g^F(s,u,w_\kappa),$$

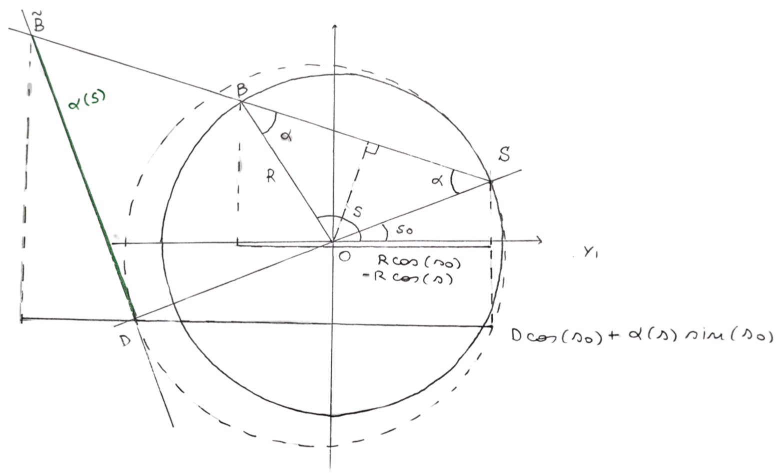
with $r\cos\alpha = \sqrt{D^2 + w_\kappa^2}$. Then in the same way as for Theorem 5 we obtain:

$$-\frac{1}{\|x - y(s)\|} g^F(s,\beta) = \frac{1}{v^*(s,x)} g^F(s,u,w_\kappa),$$

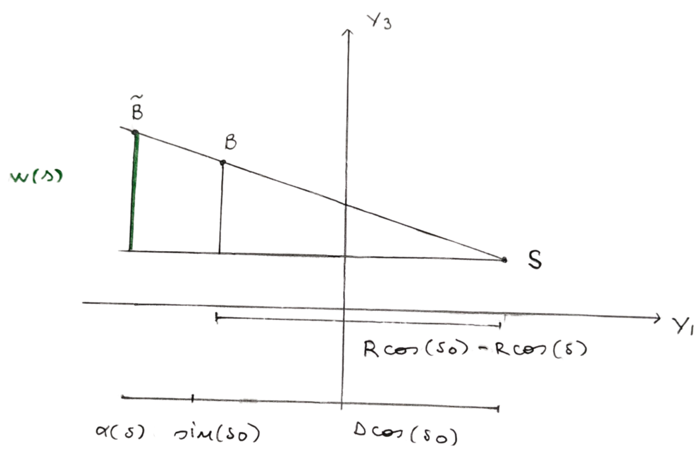
as claimed. □

Appendix C

Non Constant Tam Danielsson Window



(a) Top view, for the derivation of $\alpha(s)$.



(b) Side view, for the derivation of $w(s)$.

Figure C.1: Views of the projected trajectory of the helical locus with source at $y(s_0)$.

Appendix D

Implementation Details

While the main chapters of this thesis focus on the theoretical foundations and methodology of tomographic reconstruction, the core of the work has been the practical implementation of the methods described in Chapter 3 within the Python-based Operator Discretization Library (ODL). The full implementation developed for this project is available in the GitHub repository odl-katsevich.

This appendix is intended to guide readers who want to work with this code and perform tomographic reconstruction experiments in ODL, using either Filtered Backprojection (FBP) or Katsevich's method.

D.1 Operator-Based Setup with ODL

ODL already provides many of the core components needed for tomographic reconstruction. This section outlines how to define the reconstruction space, the acquisition geometry, and the operator chain used for filtered backprojection, which also serve as the base to implement the Katsevich reconstruction.

Defining Discretized Spaces and Acquisition Geometry

ODL defines discretized function spaces using constructors that generate either uniform or non-uniform grids over a given domain. For example:

```
reco_space = odl.uniform_discr(min_pt, max_pt, shape)
```

creates a uniformly discretized vector space over a box. This space represents the image or volume to be reconstructed. The physical extent of the domain is defined by `min_pt` and `max_pt`, while the resolution is determined by `shape`.

The acquisition geometry defines the physical layout of the scanner, including the source trajectory and the detector configuration. In this thesis, the geometry is a helical cone-beam configuration, implemented for both flat and curved detectors. This setup is instantiated using ODL's `ConeBeamGeometry` class, which supports both circular and helical source trajectories through appropriate parameter choices.

The helical geometry can be constructed as follows:

```
geometry = odl.tomo.ConeBeamGeometry(
    angle_partition, detector_partition,
    src_radius, det_radius,
    pitch=pitch, det_curvature_radius=(det_curv, 0),
    det_axes_init=[0, 1, 0],
    src_to_det_init=([1, 0, 0], [0, 0, 1])
)
```

Here, setting `pitch` to a non-zero value enables a helical rather than circular source path.

The parameter `angle_partition` defines the sampling of the rotation angle s along the helical trajectory, while `detector_partition` specifies the sampling on the detector plane, using local detector coordinates (u, w) (for flat) or (α, w) (for curved). The physical distances from the rotation axis to the source and detector are given by `src_radius` and `det_radius`, respectively.

The `pitch` parameter defines the displacement in the rotation axis direction per full rotation of the source and is specified as a scalar. This limits the geometry to a constant-pitch helix. If a variable pitch is needed, the geometry object supports the dynamic behavior using the optional arguments:

```
    src_shift_func : callable, optional
    det_shift_func : callable, optional
```

These parameters accept user-defined functions that take the rotation angle s as input and return a displacement vector for the source or detector at each angle. The returned vector should be expressed in the local coordinate system defined by:

- the source-to-detector axis ($\mathbf{e}_v(s)$),

- the axis tangential to the rotation ($\mathbf{e}_u(s)$),
- the rotation axis ($\mathbf{e}_w$).

To fully define the geometry, one must also specify the initial orientation of the detector coordinate system in the global xyz space. This is done by providing the parameters:

- `det_axes_init[0]`: $\mathbf{e}_u(0)$,
- `src_to_det_init`: $\mathbf{e}_v(0)$,
- `det_axes_init[1]`: $\mathbf{e}_w$.

By default, these are aligned with the global axes:

$$\mathbf{e}_u(0) = \mathbf{e}_x, \quad \mathbf{e}_v(0) = \mathbf{e}_y, \quad \mathbf{e}_w = \mathbf{e}_z$$

Note that under this default convention, the source trajectory is parameterized as:

$$y(s) = [-\sin s, R\cos s, p(s)]^T$$

which is effectively a counter-clockwise rotation by $-\pi/2$ with respect to the conventional parametrization used in literature, as presented in Equation (2.24). Therefore, it is important to explicitly define these orientation parameters when trying to match a given trajectory definition.

Finally, the detector curvature can be customized via the `det_curvature_radius` parameter. The default value corresponds to a flat detector. To configure a cylindrical detector, one must pass a tuple `(curv, 0)` where `curv` > 0 specifies the curvature radius along the horizontal detector direction $\mathbf{e}_u$, while zero curvature is maintained in the rotation axis direction $\mathbf{e}_w$. A typical choice for the curvature is `curv = src_radius + det_radius`.

Forward and Adjoint Ray Transform

Once the reconstruction space and geometry are defined, the forward projection operator is created using:

```
ray_trafo = odl.tomo.RayTransform(reco_space,
                                  geometry, impl='astra_cuda')
```

This operator represents the X-ray transform: it maps a 3D volume to its projections by computing line integrals along the rays defined by the geometry. The implementation is delegated to the ASTRA Toolbox, which handles memory allocation and execution on the GPU.
The adjoint operator of the ray transform, which corresponds to backprojection, is obtained via:

```
bp = ray_trafo.adjoint
```

When called with some data, this operator maps that sinogram back into the image space.

Filtered Backprojection Pipeline

The operator formulation of the FBP inversion formula is:

$$f(\cdot) = \mathcal{R}^{*}\mathcal{F}^{-1}(\hat{v}(\sigma) \cdot \mathcal{F}(\cdot))$$

ODL provides support for constructing a complete filtered backprojection (FBP) operator with the function `fbp_op` in the file

```
odl/odl/tomo/analytic/filtered_back_projection
```

This operator encapsulates the three main stages of the FBP algorithm: Fourier transform of the sinogram, frequency-domain filtering, and backprojection.

```
fbp = fbp_op(ray_trafo, filter_type='ramp', padding=True)
    = ray_trafo.adjoint * (fourier.inverse *
                           ramp_function *
                           fourier)
```

Here, `fourier` denotes the `FourierTransform` operator implemented in ODL, and `ramp_function` is the selected frequency-domain filter (*e.g.,* ramp, Hann). The result is an operator that maps sinogram data directly to a reconstructed image.

The Katsevich reconstruction method is structured to follow the FBP implementation as closely as possible. The existing `fbp_op` function can still be reused, allowing the reconstruction to be expressed as the composition of a filter and the adjoint of the ray transform:

```
kats = fbp_op(ray_trafo, filter_type='Katsevich')
= ray_trafo.adjoint_kats *
katsevich_filter_op(ray_trafo)
```

Beyond this point, the internal components of FBP are no longer applicable. In particular, the standard FBP filter `fbp_filter_op` must be replaced with a dedicated operator tailored to the Katsevich method. Details of this implementation are provided in Appendix D.2.

D.2 Implementation of Katsevich Reconstruction in ODL

This chapter presents the implementation of the Katsevich reconstruction algorithm within the ODL framework. The core idea is to construct the key components of the pipeline (filtering, windowing, and backprojection) as modular operators that conform to ODL's structure. This design allows the reconstruction method to be applied efficiently and flexibly across different acquisition geometries. The code of these operators can be found in the GitHub repository odl-katsevich in the file `odl/odl/tomo/analytic/filtered_back_projection`.

The reconstruction pipeline is therefore composed of three main stages: a filtering operator that performs Hilbert transform-based filtering along re-parametrized lines, the application of the TD window as a masking operator on the sinogram, and a backprojection operator that maps the filtered data into the image domain. Each of these components is implemented as an ODL-compatible operator and supports both constant and variable pitch configurations, with optimizations when the pitch is constant. The following sections describe the implementation of each component in detail.

D.2.1 Filter Operator

We define a custom filtering operator, `KatsevichFilterCurved` and `KatsevichFilterFlat`, which encapsulates all the geometry-dependent rebinning and filtering steps described in the algorithm. This operator is implemented through a class-based structure that supports both flat and curved detector geometries, sharing the same logic. Upon initialization, the operator extracts all relevant parameters from the `ray_trafo`, including the detector shape and coordinates, projection angles, and source trajectory. A key part of the setup is determining whether the pitch is constant or varies across projections. This is done by evaluating the source and detector shift functions provided by the geometry: if these are identically zero over all projection angles, the pitch is considered constant. Otherwise, the operator assumes a variable-pitch trajectory. This distinction allows for significant computational optimization, since in the constant pitch case some quantities involved in the filtering process are independent of the projection angle and can therefore be precomputed once.

Based on the pitch classification, the operator precomputes during initialization all geometry-dependent quantities needed in the filtering process. In particular, two key mappings are computed in this phase: the first, `fwd_rebin_w`, stores for each filtering line the corresponding detector vertical coordinate at which to sample during forward rebinning. The second mapping supports backward rebinning: the array `rebin_row` stores, for each detector element, the index of the two filtering lines that bracket it, while `rebin_fracs_0` and `rebin_fracs_1` store the interpolation weights $c(\alpha, w, \ell)$ and $1 - c(\alpha, w, \ell)$ used to map filtered values back onto the detector grid presented in Section 3.1.1. These arrays are either shared across all views (in the constant pitch case) or stored per view (in the variable pitch case), and allow the filtering steps to be executed without recomputing geometry-specific information at runtime.

When the operator is later called with projection data, the filtering pipeline is applied as a sequence of member functions, each corresponding to one of the algorithmic steps described in Section 3.1.1 and in Section 3.1.2. Here we will discuss the implementation for the curved detector setup; the logic for the flat detector is analogous, with the exception that the cosine weighting step, at the end of the filtering, is specific to the curved detector and does not appear in the flat case.

The pipeline begins with a directional derivative of the sinogram, implemented in the two variants presented in Section 3.1.1 `cf1_derivative_NDH` and `cf1_derivative_K`, which use finite differences to approximate the derivative along the helical trajectory. The result is then modulated by a geometrical length weighting in `cf2_length_weighting`, applied pointwise as a function of the detector coordinate.

The weighted sinogram is rebinned onto virtual filtering lines using `cf3_forward_rebin` or `cf3_forward_rebin_nonconst_pitch`, which interpolate the detector grid data on the filtering lines using the precomputed array `fwd_rebin_w`. The filtering itself is carried out via a discrete Hilbert transform in `cf4_hilbert_transform`, implemented as a convolution along detector columns using a windowed kernel constructed at runtime.

After filtering, the result is mapped back onto the original detector grid through backward rebinning, using either `cf5_backward_rebin` or `cf5_backward_rebin_nonconst_pitch`, which apply linear interpolation using the arrays `rebin_row`, `rebin_fracs_0`, and `rebin_fracs_1`. Finally, a cosine weighting is applied in `cf6_cosine_weighting` to adjust the projection density along the fan-beam.

Each of these steps operates exclusively on the data provided at runtime, and rely on the geometric mappings precomputed during the initialization. This clear separation between geometry-dependent and data-dependent computations makes the filtering operator efficient in execution and reusable across different sinograms, without the need to recompute the most computationally intensive part of the algorithm, *i.e.,* the forward and backward rebinning maps. This aspect will be further discussed in Chapter 6.

As a final note, while the general implementation supports both flat and curved detector geometries, the focus of this work was on curved detectors, the standard detector used in clinical CT systems. In particular, support for general variable-pitch trajectories is implemented only in the curved case. The flat detector implementation is kept as a simplified case used primarily for validation and benchmarking purposes.

D.2.2 Backprojection Operator

The full backprojection is given by the composition of the TD window on the filtered projection data and the adjoint ray transform operator:

```
ray_trafo.adjoint ( td_window(ray_trafo)  * proj_data )
```

TD Windowing

The Tam–Danielsson (TD) window is implemented as a mask applied to the projection data, suppressing values outside the region bounded by the projections of adjacent helical turns on the detector. The function `td_window_curved` computes and applies this mask pointwise to the sinogram.

In the constant pitch case, the upper and lower detector bounds `w_top` and `w_bottom` remain fixed across views and are precomputed from the acquisition parameters (source radius, detector radius, pitch) using the analytical formulas 3.12, 3.13. The mask consists of a central region set to one, with linear ramps (controlled by a smoothing parameter `a`) fading the mask to zero. This 2D mask is then broadcast over all projection angles.

With variable pitch, the mask must be adapted to the changing geometry. For each view k, the detector coordinates of the projections of the helical turns above and below are computed via equations 3.14 and 3.15. After interpolation to align the values on the parametrizes curve with the detector grid, the corresponding mask is built and applied individually. While more computationally expensive, this approach ensures that the TD window remains geometrically consistent with the helical path, even under nonuniform pitch.

Weighted Adjoint

A key feature of the Katsevich backprojection formula is the presence of a voxel-dependent weighting term, given by:

$$\frac{1}{v^*(s,\mathbf{x})} = \frac{1}{R - \langle \mathbf{x}, \mathbf{e}_v(s) \rangle}, \tag{D.1}$$

where R is the distance source to origin. This weight depends simultaneously on the reconstruction point $\mathbf{x}$ and on the source position along the trajectory, parameterized by the rotation angle s. In contrast to standard filtered backprojection, where backprojection and weighting are typically decoupled, this formula requires the weighting to be applied inside the integral 2.34, which makes it incompatible with standard implementations that treat backprojection as a global operator.

It was initially investigated whether this type of backprojection could still be implemented using the standard `ray_trafo.adjoint` method provided by ODL. In principle, this would require an ASTRA binding capable of accepting a user-defined weighting function and applying it directly during the backprojection process. Unfortunately, such functionality is not currently available in the ASTRA Toolbox, which made it impossible to directly implement the exact Katsevich backprojection in this way.

As a result, two alternative strategies were considered to incorporate the voxel- and angle-dependent weighting required by the Katsevich formula:

Implementation of a Custom Adjoint Operator. The first approach consisted in designing a custom operator that performs the standard ASTRA-based backprojection and subsequently applies the voxel-wise weights as a post-processing step. To this end, a new method, `ray_trafo.adjoint_kats`, was implemented. This method follows the ODL interface and internally executes the backprojection using ASTRA, but then switches to the *CuPy* backend to apply the scaling factor defined in Equation (D.1) on the resulting volume. The weight is evaluated at each voxel for the corresponding angle s, and the result is multiplied element-wise with the backprojected data. This method was largely inspired by the implementation in [17].

While functional, this method is not ideal for two reasons. First, *CuPy* is not supported in ODL yet. Second, the post-processing approach breaks the clean operator-based structure that characterizes ODL.

View-Wise Backprojection The second strategy preserves the standard ODL operator interface while enabling voxel-dependent weighting. It consists in decomposing the full backprojection into a sequence of per-view operations. In particular, for each source position corresponding to a fixed angle s_k, a new `RayTransform` operator is created using a geometry object whose angle partition contains only that single angle:

```
ray_trafo_k = RayTransform(space, geometry_k)
```

The geometry `geometry_k` is a copy of the original acquisition geometry but restricted to the angle s_k. The corresponding adjoint operator `ray_trafo_k.adjoint` then performs standard ASTRA-based backprojection using only the projections from the single source position s_k. After this backprojection step, the weighting function in Equation (D.1) is evaluated for the fixed s_k and applied to the resulting volume slice.

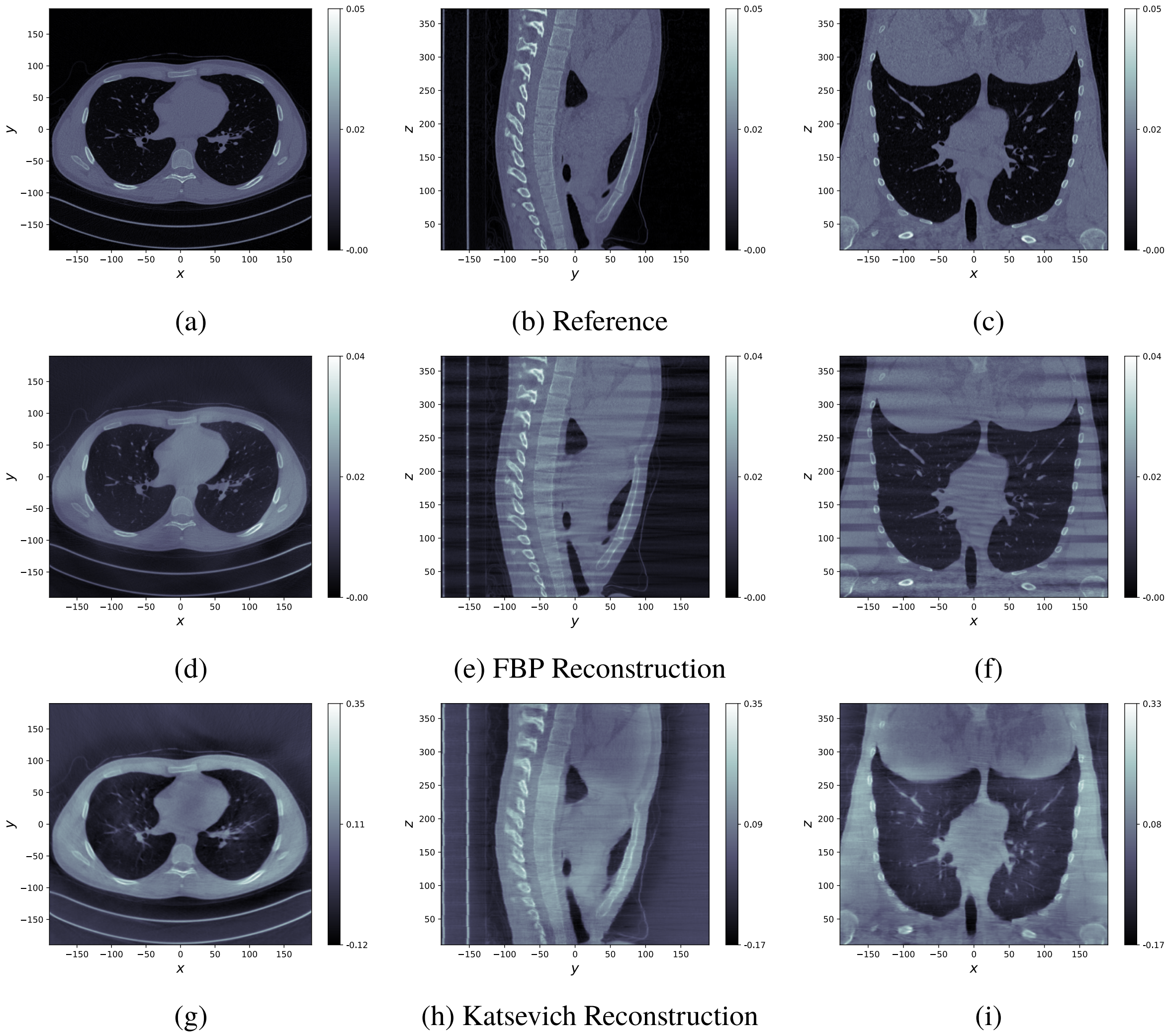
This procedure is repeated for each source angle in the acquisition trajectory. The weighted contributions from each backprojected slice are accumulated into the final reconstructed volume.

Although more computationally demanding due to the need to instantiate many temporary geometry objects and run multiple backprojections, this method maintains full compatibility with ODL's operator framework and relies on standard methods already implemented and the results should remain unaffected. Unfortunately, this method wasn't completed within the time dedicated to this thesis and therefore won't be present in the results. Its implementation will be left as a future work to be implemented.

Appendix E

Extra Mayo Dataset Results

All SDGs are within the range [1-17].



(a) (b) Reference (c)

(d) (e) FBP Reconstruction (f)

(g) (h) Katsevich Reconstruction (i)

Figure E.1: Axial, sagittal and coronal views of the reconstructions of the chest scan C004 in the Mayo dataset using FBP E.1d, E.1e, E.1f and Katsevich E.1g, E.1h, E.1i. The reference reconstruction E.1a, E.1b, E.1c is provided by the Mayo dataset and is executed using an FBP-type method. Obtained from a sinogram of 13782 projections.